

MUSIC LESSONS

Grade 2 — Listen and Play

Keyboard · Guitar

For Students Age 8 to Adult

25 Lessons · Tune · Rhythm · Scales · Chords

Simple English. Deep Knowledge. Real Music.

Barnabas Thalam

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MUSIC THEORY REFERENCE

📖 How to Use This Book

Welcome, music student! This book is your guide for Grade 2 — Listen and Play. It is written in simple English so that every student, whether you are 8 years old or an adult, can understand and enjoy music.

This book covers four big topics:

- Tune — The melody of a song, the main notes you hear and sing.
- Rhythm — The timing and beats that make music move.
- Scales — The set of notes that give music its feeling (happy, sad, exciting).
- Chords — Groups of notes played together to support the melody.

Before every lesson:

1. Open YouTube and find any song you like.
2. Press play and just listen carefully.
3. Then open this book and follow the lesson.

Each lesson has: a Definition, an Explanation, a Formula (when needed), Examples, Workings, and Practice Activities. Follow each section step by step. Do not rush. Music is a journey — enjoy every note!

> 🎸 **For Guitar Students**
> All lessons apply to guitar. Wherever 'keyboard' or 'keys' are mentioned, think about your guitar strings and frets. Notes, tones, and pitches work the same way on both instruments.

PART 1 — OBSERVE THE TUNE

Note: Play any song on YouTube and observe these things.

Lesson 1 — Learn and Match the Tones

What is a Tone?

A tone is a musical sound that has a specific pitch. Pitch means how high or low a sound is. When you pluck a guitar string or press a piano key, you create a tone.

Every musical note is a tone. The basic music notes are:

Note Name	How It Sounds
C	Calm, like a home base note
D	Slightly higher, bright
E	Warm, gentle feeling
F	A small step up from E
G	Open, full, easy to hear
A	Bright and clear
B	Wants to go back to C

What is 'Matching' a Tone?

Matching a tone means you hear a sound in a song and then you find that exact same note on your instrument. If a song plays a 'G', you must also play a 'G'. This is called ear training.

How to Match a Tone — Step by Step

1. Listen to one note from the song. Try to hear it clearly.
2. Sing or hum that note. Yes, hum it! This trains your ear.
3. Now try notes on your keyboard/guitar one by one.
4. When the note on your instrument sounds the same as what you hummed, you have matched it!
5. Write down the note name.

> 🎵 **Example — Matching a Tone**

You play 'Happy Birthday'. The first sound is a 'G'. Press G on your keyboard and hum it. Now press each key near that area. When they match your hum, that is your note.

✏️ PRACTICE ACTIVITIES

1. Play any YouTube song. Pause after the very first note. Hum that note. Find it on your keyboard or guitar.
2. Play each white key on the keyboard slowly — C, D, E, F, G, A, B. Hum each one. Notice the differences.
3. Ask a friend or teacher to play a note. Without looking, try to find that note on your instrument.
4. Play 5 songs on YouTube. Write down what you think the first note is for each song.

Lesson 2 — Find the Starting Pitch of the Tune

What is Pitch?

Pitch is how high or low a note sounds. A bird chirp is high pitch. A big drum is low pitch. Every song starts on one specific pitch — this is called the Starting Pitch or Root Note.

> 📌 **Why is Starting Pitch Important?**
> If you find the starting pitch of a song, you can play the whole tune by following the other notes from that starting point. It is like finding the first word of a sentence — once you have it, the rest follows.

Formula — How to Find the Starting Pitch

Listen → Hum the first note → Test keys/strings → Confirm → Write it down

Detailed Steps

1. Play the song on YouTube. Listen to the very first vocal or melody note.
2. Pause the song immediately after the first note.
3. Hum or sing that note out loud. Keep humming.
4. Go to your keyboard and start from middle C (the C in the middle of the keyboard).
5. Press each key and listen. Does it match your hum?
6. Move up (higher) or down (lower) until you find the exact match.
7. When it matches — congratulations! You found the starting pitch.

Keyboard Guide — Finding Middle C

On a standard keyboard, the Middle C is the C note closest to the center of the keyboard. It is usually marked with a sticker or it is the 4th C from the left. On guitar, the Middle C is on the 2nd string (B string), 1st fret, or 3rd string (G string), 5th fret.

> 🎵 **Example — 'Twinkle Twinkle Little Star'**
> The first note is 'C'. When you press C on the keyboard and hum the starting note of Twinkle Twinkle, they will match perfectly. Try it!

> 🎵 **Example — 'Let It Be' by The Beatles**

The first melody note in the verses starts around 'C' or 'E' depending on the key. Listen carefully and find which one matches.

✏️ PRACTICE ACTIVITIES

1. Play 'Twinkle Twinkle Little Star' on YouTube. Find the starting pitch and write it down.
2. Play 'Happy Birthday' on YouTube. Hum the first note. Find it on your instrument.
3. Choose any 3 songs from YouTube. Find and write the starting pitch of each one.
4. Practice humming first, then finding the note. Time yourself — try to find it in under 30 seconds.

Lesson 3 — Follow the Next Pitches

What Does 'Following the Pitches' Mean?

After you find the starting pitch, the tune moves to other pitches. Some notes go UP (higher) and some go DOWN (lower). Following the pitches means you listen to each note and play it in the correct order on your instrument.

Understanding Movement Between Pitches

Notes in music move in two main ways:

- Step: Moving to the very next note. Example: C → D (one step up).
- Jump/Leap: Skipping one or more notes. Example: C → E (skips D) or C → G (bigger jump).

Type of Movement	Example
Step Up	C → D, D → E, E → F
Step Down	G → F, E → D
Small Jump Up	C → E, D → F, G → B
Big Jump Up	C → G, D → A
Jump Down	G → D, A → E

Formula — Following the Tune

First Note → Is next note higher or lower? → By how much? → Play it → Continue

How to Follow Pitches in Practice

1. Find the starting pitch (Lesson 2).
2. Listen to the second note. Ask yourself: Is it higher or lower?
3. If higher — move your finger to the right on keyboard, or up the frets on guitar.
4. If lower — move your finger to the left on keyboard, or down the frets.
5. Start with just 3 notes. Then add more one by one.
6. Play the notes you found in order — you are now playing the melody!

🎵 **Example — Twinkle Twinkle Little Star (First Line)**

The first line goes: C C G G A A G (rest) F F E E D D CC to C = same note. C to G = jump up. G to A = step up. A back to G = step down. Then F, E, D, C going down.

✏️ PRACTICE ACTIVITIES

1. Listen to 'Mary Had a Little Lamb'. Follow the first 7 notes and write if each note is UP, DOWN, or SAME.
2. Play Twinkle Twinkle — start from C and follow: C C G G A A G.
3. Choose any slow song on YouTube. Pause after every 3 notes and find them on your instrument.
4. Practice singing the direction of notes: say 'UP' or 'DOWN' out loud as the song plays.

Lesson 4 — Observe the Speed, Slow, Gaps, and Bends of Tunes

Introduction

Music is not just about which notes to play — it is also about HOW you play them. The same notes played a fast or slow sound completely different. This lesson teaches you to hear and copy the feel of a melody.

Speed (Tempo)

Tempo is the speed of the music. It tells you how fast or slow to play the notes.

Tempo Word	Meaning
Largo	Very very slow
Adagio	Slow and gentle
Andante	Walking speed (medium)
Moderato	Moderate — not too fast, not too slow
Allegro	Fast and lively
Presto	Very fast

Slowing Down and Speeding Up

- Rallentando (rall.) — Gradually slow down towards the end.
- Accelerando (accel.) — Gradually get faster.
- Ritardando (rit.) — Slow down, often at the end of a section.

Gaps (Rests)

A rest is a silence in music. It is a gap where no note is played. Rests are just as important as notes! They give music breathing space.

A song without rests is like a sentence with no spaces between words — it makes no sense!

Bends

A bend is when a note slides smoothly up or down to another note without a gap. It is very common in guitar playing and in vocal music. On guitar, you push the string sideways while holding a fret to make the pitch rise smoothly. On keyboard, you use a pitch bend wheel.

🎵 **Example — Observing a Song**

Play 'Someone Like You' by Adele on YouTube.Speed: Slow and emotional (Andante)Gaps: Listen for small pauses between phrasesBends: Notice how her voice slides between some notes

✏️ **PRACTICE ACTIVITIES**

1. Play a fast song (like 'Uptown Funk') and a slow song (like 'Hallelujah'). Describe the tempo of each.
2. Listen to a song and clap every time there is a silence (rest). Count how many rests you hear.
3. If you have a guitar, practice one note bend on the 3rd fret of the G string.
4. Record yourself humming a song. Listen back and notice where your voice naturally speeds up, slows down, or bends.

Lesson 5 — Learn to Write the Tones of the Tunes

Why Write Down Notes?

When you write notes down, you never forget the tune. Writing notes is like writing words — once you learn the system, you can read any music and remember any song.

Two Ways to Write Notes

- Number System (Nashville Number System / Sargam)
- Letter Name System (C D E F G A B)

The Number System

This system uses numbers 1 to 7 for the seven notes of a scale. If your song is in C major:

Number	Note in C Major
1	C
2	D
3	E
4	F
5	G
6	A
7	B

So if a tune goes: C E G G E C — you write it as: 1 3 5 5 3 1

The Letter Name System

Simply write the letter names of the notes in order. Add a dash (-) for a long note, and a dot (.) for a short note or a rest.

Twinkle Twinkle: C C G G A A G - F F E E D D C -

Arrows for Direction

Use arrows to show if notes go up or down:

- ↑ = note goes UP
- ↓ = note goes DOWN
- → = note stays the same

🎵 **Example — Writing Twinkle Twinkle**

First line: C C G G A A GUsing numbers (C major scale): 1 1 5 5 6 6 5Using letters with direction: C→ G↑ G→ A↑ A→ G↓

✏️ **PRACTICE ACTIVITIES**

Listen to 'Happy Birthday'. Write out the first line using letter names: G G A G C B.

Convert 'Twinkle Twinkle' (C C G G A A G) to number notation.

Choose any children's song on YouTube. Write the first 8 notes using letter names.

Write 'Mary Had a Little Lamb' in both letter names AND number notation.

PART 2 — RHYTHM

Rhythm is the heartbeat of music.

Lesson 6 — Counts and Timings

What is Rhythm?

Rhythm is the pattern of beats in music. A beat is a steady pulse — like the ticking of a clock or your heartbeat. Rhythm tells you WHEN to play each note.

What is a Count?

In music, we count beats in groups. The most common groups are 2, 3, or 4 beats. Each group is called a MEASURE or BAR.

1 Bar = A fixed number of beats counted together (e.g., 1-2-3-4)

Note Values — How Long Each Note Lasts

Note Name	How Many Beats
Whole Note	4 beats — count: 1-2-3-4
Half Note	2 beats — count: 1-2
Quarter Note	1 beat — count: 1
Eighth Note	Half a beat — count: 1-and
Sixteenth Note	Quarter of a beat — count: 1-e-and-a

Rest Values — Silence

Rest Name	Silence For
Whole Rest	4 beats of silence
Half Rest	2 beats of silence
Quarter Rest	1 beat of silence
Eighth Rest	Half a beat of silence

The Formula for Counting

Count out loud: 1 - 2 - 3 - 4 | 1 - 2 - 3 - 4 | (always steady, like a clock)

✏️ PRACTICE ACTIVITIES

Clap your hands in a steady beat for 1 minute. Keep it the same speed all the time.

Listen to any song on YouTube. Try to count along: 1-2-3-4, 1-2-3-4...

Tap the table with 4 equal taps, then stop (that is one bar). Repeat 8 times.

Play a whole note C on keyboard: press and hold for 4 beats. Play a quarter note C: short press, 1 beat.

Lesson 7 — Timing to Start the First Note of the Tune

Introduction

Not all songs start on beat 1. Some songs start on beat 2, beat 3, or even on an 'and' between beats. This first note is very important — if you start at the wrong time, the whole song feels wrong.

What is the Anacrusis (Pickup / Upbeat)?

Sometimes a song starts before beat 1 — this is called an anacrusis or pickup note. It is like a little running start before the main beat begins.

> 🎵 **Example — 'Happy Birthday'**
> 'Happy Birthday to you' — the word 'Hap-' is a pickup note. It comes BEFORE beat 1. Beat 1 lands on 'Birth-'.

How to Find When to Start

1. Listen to the song with counting: 1-2-3-4, 1-2-3-4...
2. Find out when the main beat (beat 1) of each bar falls — it is usually where the music feels 'heavy' or 'strong'.
3. Notice where the melody starts — is it on beat 1? Or is it earlier (pickup) or later?
4. Practice clapping the beat, then add the melody on top.

Beat 1 = the strongest beat in a bar. Melody can start on any beat or even before beat 1.

Song Example	Starting Beat
Twinkle Twinkle	Starts on Beat 1
Happy Birthday	Starts on the 'and' before beat 1 (pickup)
We Will Rock You	Starts on beat 1 after 2 bars of drum intro
Amazing Grace	Starts on beat 3 (pickup to bar 1)

✏️ **PRACTICE ACTIVITIES**

GRADE 2 : Listen and Play

1. Play 'Twinkle Twinkle'. Count 1-2-3-4 and play C on beat 1.
2. Play 'Happy Birthday'. Find the pickup note before beat 1 and practice starting there.
3. Listen to 3 songs. For each one, identify which beat the melody starts on.
4. Practice tapping your foot on beat 1 while you listen to music. Only tap on beat 1.

Lesson 8 — Timing to Follow the Tunes as It Is

What Does 'Following the Tune as It Is' Mean?

Once you find the notes and the starting beat, the next challenge is to keep every note at exactly the right time — just like the original song. This is called rhythmic accuracy.

Short Notes vs Long Notes

As you follow the melody, some notes are held for a long time and some are very short. You must match the exact length of each note.

Long Note = Hold the key/string down. Short Note = Quick press and release.

Using a Metronome

A metronome is a tool (or app) that beeps at a steady speed to help you keep time. It is the most important tool for practising timing.

- Free apps: 'Metronome Beats' on iPhone/Android
- Website: metronome-online.com
- Start slow! Set the metronome to 60 BPM (beats per minute) and learn the song slowly first.

How to Use a Metronome

1. Set a slow speed (60 BPM).
2. Count: 1-2-3-4 with the metronome clicks before you start.
3. Play your notes in time with the clicks.
4. When it feels easy, increase the speed by 5 BPM.
5. Keep increasing until you reach the song's original speed.

🎵 **Working Example — Following Twinkle Twinkle**

Metronome at 80 BPM. Each click = 1 beat.Beat 1: C | Beat 2: C | Beat 3: G | Beat 4: GBeat 1: A | Beat 2: A | Beat 3-4: G (hold 2 beats)Keep this exact timing for every line.

✏️ **PRACTICE ACTIVITIES**

1. Download a metronome app. Set it to 60 BPM. Clap on every beat for 1 minute.
2. Play Twinkle Twinkle with a metronome at 60 BPM. One note per beat.
3. Record yourself playing a song, then listen back. Where did you rush or slow down?
4. Play any song at half speed first (slow). Then gradually increase the tempo.

Lesson 9 — Timing of Rolling

What is Rolling in Music?

Rolling means playing notes very quickly one after another, often going up or down a scale or chord. It creates a smooth, flowing, wave-like sound. Rolling is also called a 'run' or 'glissando' in some styles.

Types of Rolling

- Scale Roll: Playing notes of a scale quickly — C D E F G A B C (going up fast).
- Chord Roll (Arpeggio): Playing the notes of a chord one by one quickly instead of all together.
- Glissando: Sliding your finger across many keys or strings smoothly.

Formula for an Arpeggio Roll

Chord (C-E-G) → Play C then E then G one by one, very quickly in sequence

How Timing Works in Rolling

1. Each note in a roll is usually very short — often sixteenth notes or even faster.
2. The roll starts on the beat, and the notes arrive so quickly that your ear hears them almost as one event.
3. The last note of the roll usually lands on the beat.

🎵 **Example — Piano Roll on C Major Chord**

C-E-G played as: C (very fast) → E (very fast) → G (lands on beat 1).You hear three notes but they all belong to the same beat.

🎵 **Example — Guitar Arpeggio**

On a C chord (C-E-G), instead of strumming all strings at once, pick each string: E string → A string → D string → G string, all in one beat.

✏️ **PRACTICE ACTIVITIES**

1. Play C D E F G A B C as fast as you can while still being clean. That is a scale roll.

2. Play the notes of a C chord (C-E-G) one by one, slowly at first, then faster.
3. On the keyboard, try a glissando: slide your finger across all the white keys from C to C.
4. Listen to a classical piano song (e.g., Beethoven's 'Moonlight Sonata'). Identify the arpeggio rolls.

Lesson 10 — Timing of Intro and Ending

What is an Intro?

An intro (introduction) is the musical part at the BEGINNING of a song, before the singing or main melody starts. The intro sets the mood and tells the listener what kind of song is coming.

What is an Ending (Outro)?

An ending or outro is the final section of a song. It signals that the song is finishing. The music often slows down, gets quieter, or repeats a phrase fading out.

Types of Intros

Intro Type	Description
Chord Intro	Chords played before the melody starts. Common in pop songs.
Riff Intro	A catchy repeated melody phrase before the song begins (e.g., 'Smoke on the Water').
Drum Intro	Only the rhythm section plays first (e.g., 'We Will Rock You').
Solo Intro	One instrument plays alone to open the song.

Types of Endings

Ending Type	Description
Fade Out	Song gradually gets quieter until silence. Common in recorded music.
Final Chord	Song ends on one strong chord — BANG — silence.
Ritardando End	Song slows down as it finishes.
Repeat & Fade	Last phrase repeats over and over, getting softer each time.

Intro sets the KEY, TEMPO, and MOOD. Ending brings the RESOLUTION (feeling of finish).

✏️ PRACTICE ACTIVITIES

1. Listen to 5 songs. For each, describe the intro: how many beats? What instrument leads?
2. Play a 4-bar chord intro using C - Am - F - G (one chord per bar).
3. Practice an ending: play C major chord, then slow down and hold the final note.
4. Create your own intro for a simple song — 4 bars, any notes you like.

Lesson 11 — 2/4 Time Signature

What is a Time Signature?

A time signature is like a fraction that appears at the beginning of a piece of music. It tells you two things:

- Top number: How many beats are in each bar.
- Bottom number: What type of note gets one beat.

Time Signature = Top Number / Bottom Number2/4 = 2 beats per bar, quarter note gets 1 beat

Understanding 2/4 Time

In 2/4 time, there are 2 beats per bar. You count: 1-2 | 1-2 | 1-2...

Beat 1 is STRONG (heavy). Beat 2 is WEAK (light).

Beat	Feel
Beat 1	STRONG — where you feel the pulse most
Beat 2	weak — lighter, like a small bounce

What Music Uses 2/4?

- March music (soldiers marching: LEFT - right - LEFT - right)
- Polka music
- Many folk songs

🎵 **Example — Marching in 2/4**

Count: 1-2 | 1-2 | 1-2Left foot = Beat 1 (strong). Right foot = Beat 2 (weak).'Star Wars' march theme is in 4/4 but march feel — similar concept.

🎵 **Example Song — 'Twinkle Twinkle' (can be in 2/4)**

Bar 1: C C | Bar 2: G G | Bar 3: A A | Bar 4: G -

How to Clap 2/4

1. CLAP hard on beat 1.
2. tap lightly on beat 2.
3. Repeat: CLAP-tap, CLAP-tap...

✏️ PRACTICE ACTIVITIES

1. Clap 2/4 time: CLAP-tap, CLAP-tap. Do this for 2 minutes with a metronome at 80 BPM.
2. Play C on beat 1 and G on beat 2. Repeat for 4 bars. That is 2/4 time.
3. Listen to a march on YouTube. Try to count the 1-2, 1-2 beat.
4. Compose a 4-bar melody in 2/4 time. Use notes C D E G. Write it down.

Lesson 12 — 3/4 Time Signature

What is 3/4 Time?

In 3/4 time, there are 3 beats per bar. Count: 1-2-3 | 1-2-3 | 1-2-3...

Beat 1 is STRONG. Beats 2 and 3 are weak.

3/4 → Count: 1 - 2 - 3 | 1 - 2 - 3 | (3 beats per bar, quarter note = 1 beat)

The 'Waltz' Feel

3/4 time creates a swinging, dance-like feel. This is why it is also called 'Waltz time.' When you waltz, you step on 3 beats: STEP-2-3, STEP-2-3.

Beat	Feel
Beat 1	STRONG — step/down
Beat 2	weak — floating
Beat 3	weak — floating

🎵 **Example Songs in 3/4**

'Happy Birthday' — HAPpy BIRTHday TO you = 1-2-3 feel'My Favorite Things' (Sound of Music)'Moonlight Sonata' — Beethoven'Take Me Out to the Ballgame'

Happy Birthday in 3/4

Hap - py | Birth - day to | You ... | Hap - py | 1 - 2 | 3 - 1 - 2 | 3 ... | 1 - 2 |

Clapping 3/4

1. CLAP hard on beat 1.
2. tap lightly on beat 2.
3. tap lightly on beat 3.
4. Repeat: CLAP-tap-tap, CLAP-tap-tap.

✏️ **PRACTICE ACTIVITIES**

1. Sing Happy Birthday while tapping: CLAP-tap-tap, CLAP-tap-tap. Feel the 3/4.
2. Play C-E-G one note per beat for 3 beats. Rest on beat 1 of the next bar. That is one 3/4 pattern.
3. Find 2 songs on YouTube in 3/4 time. Listen for the 1-2-3, 1-2-3 feel.
4. Create a 4-bar melody in 3/4 time using notes C E G A. Write it down.

Lesson 13 — 4/4 Time Signature

What is 4/4 Time?

4/4 is the most common time signature in popular music. There are 4 beats per bar. It is so common it is called 'Common Time' and is sometimes written as a letter 'C' instead of 4/4.

4/4 → Count: 1 - 2 - 3 - 4 | 1 - 2 - 3 - 4 | (4 beats per bar)

Strong and Weak Beats in 4/4

Beat	Feel
Beat 1	STRONGEST — the main accent
Beat 2	weak
Beat 3	MEDIUM strong — secondary accent
Beat 4	weak — anticipates beat 1

In pop music, beats 2 and 4 often have a snare drum hit — this creates the 'backbeat' that makes you want to move.

🎵 **Example Songs in 4/4**

Almost ALL pop songs: 'Shape of You', 'Blinding Lights', 'Bohemian Rhapsody"Let It Be' - Beatles'Twinkle Twinkle Little Star"Jingle Bells'

Building a 4/4 Pattern

The most basic guitar or keyboard rhythm in 4/4:

Bar = [Note/Chord] -- play on beat 1 and hold for 4 beats, ORBar = [1] [2] [3] [4] -- play a note/chord on every beat

Common 4/4 Rhythmic Patterns

Pattern Name	How to Clap or Play
Basic 4	Clap on every beat: 1 2 3 4
On 2 and 4	Clap only on beats 2 and 4 (backbeat)
On 1 and 3	Clap only on beats 1 and 3
Eighth note groove	Clap on all 8 eighth notes: 1-and-2-and-3-and-4-and

✏️ PRACTICE ACTIVITIES

1. Clap 4/4 time for 2 minutes: 1-2-3-4, 1-2-3-4. Use a metronome at 80 BPM.
2. Play one chord (C major: C-E-G) per beat for 4 beats. Repeat for 4 bars.
3. Listen to 'Let It Be'. Count the 1-2-3-4 and clap on 2 and 4 (backbeat).
4. Create a 4-bar melody in 4/4 using any notes. Write it down with beat numbers.

Lesson 14 — 6/8 Time Signature

What is 6/8 Time?

6/8 time has 6 beats per bar, and each beat is an eighth note. But here is the special thing: in 6/8, the beats are grouped in TWOS of three. So you feel 2 BIG beats, each divided into 3 smaller beats.

6/8 → 6 eighth notes per bar → Grouped as: (1-2-3) (4-5-6) → Feel 2 main beats

How to Count 6/8

Count: 1-2-3-4-5-6. But feel it as: ONE-two-three-FOUR-five-six.

ONE and FOUR are the strong beats. The others are light.

The 'Compound' Feeling

6/8 has a rolling, triplet-like feeling. It sounds like: DA-da-da-DA-da-da. This is also called 'compound duple' time because there are 2 main beats, each divided into 3 parts.

🎵 **Example Songs in 6/8**

'Nothing Else Matters' by Metallica'We Are the Champions' by Queen'House of the Rising Sun' by The Animals'Greensleeves' — traditional English song'When the Saints Go Marching In'

Feature	3/4 vs 6/8
Total beats per bar	3 (in 3/4) vs 6 (in 6/8)
Feel	3 equal beats vs 2 big beats with 3 sub-beats each
Count	1-2-3 vs 1-2-3-4-5-6
Common in	Waltz vs Celtic, rock ballads

✏️ **PRACTICE ACTIVITIES**

1. Clap 6/8: CLAP-tap-tap-CLAP-tap-tap. Feel the 2 big beats.

2. Listen to 'Nothing Else Matters'. Count 1-2-3-4-5-6 along with the guitar intro.
3. Play C on beat 1, E on beat 4. Rest on all other beats. That is a basic 6/8 pattern.
4. Compare: Play a melody in 3/4 and then in 6/8. Notice how the feeling changes.

PART 3 — SCALES

Scales give music its colour and emotion.

Lesson 15 — Understanding Scales

What is a Scale?

A scale is a set of musical notes arranged in order from low to high (or high to low). Think of it as a musical alphabet for a specific key. Just as words are built from letters of the alphabet, melodies are built from the notes of a scale.

The Major Scale

The major scale is the most common scale. It has 7 notes (plus the octave — the same note but higher). The major scale sounds HAPPY, BRIGHT, and CONFIDENT.

Major Scale Formula: W - W - H - W - W - W - H(W = Whole Step, H = Half Step)

What are Whole Steps and Half Steps?

- Half Step (H) = Moving to the very next key (including black keys). Example: C to C# (one semitone).
- Whole Step (W) = Moving two half steps. Example: C to D (skip C#).

Building C Major Scale

Start on C. Apply the formula: W W H W W W H

Step	Note Reached
Start	C
Whole Step up	D
Whole Step up	E
Half Step up	F
Whole Step up	G
Whole Step up	A
Whole Step up	B
Half Step up	C (octave)

The Minor Scale (Natural Minor)

The minor scale sounds SAD, DARK, and MYSTERIOUS. It uses a different formula:

Minor Scale Formula: W - H - W - W - H - W - W

🎵 **A Minor Scale**

A B C D E F G A — these are the same notes as C Major but starting from A. This is called the Relative Minor.

✏️ **PRACTICE ACTIVITIES**

1. Play the C Major scale on keyboard: C D E F G A B C. Say each note name as you play.
2. Play the A Minor scale: A B C D E F G A. Notice how it sounds sad compared to C Major.
3. Build the G Major scale using the formula (W W H W W W H). Write the notes.
4. Play each scale 5 times a day. Start slow, then faster.

Lesson 16 — Finding Scales by Using Sharps and Flats

What are Sharps and Flats?

A Sharp (#) raises a note by a half step. A Flat (b) lowers a note by a half step. On the keyboard, sharps and flats are the BLACK keys.

- C# = C sharp (one half step above C, the black key just right of C)
- Bb = B flat (one half step below B, the black key just left of B)

The Circle of Fifths

The Circle of Fifths is a diagram that shows all 12 major scales and how many sharps or flats each one has. Moving clockwise adds one sharp. Moving counter-clockwise adds one flat.

Scale	Sharps or Flats
C Major	No sharps, no flats
G Major	1 sharp: F#
D Major	2 sharps: F#, C#
A Major	3 sharps: F#, C#, G#
E Major	4 sharps: F#, C#, G#, D#
F Major	1 flat: Bb
Bb Major	2 flats: Bb, Eb
Eb Major	3 flats: Bb, Eb, Ab

How to Build a Scale Using Sharps

Apply the Major Scale formula (W W H W W W H) from any starting note. Wherever the formula requires a whole step but two adjacent keys are only a half step apart, you must use the sharp of that key.

🎵 **Building G Major Scale**

G A B C D E F# GThe 7th note (F) must be raised to F# to complete the whole step pattern.

🎵 **Building F Major Scale**

F G A Bb C D E FThe 4th note (B) must be lowered to Bb to complete the half step pattern.

✏️ PRACTICE ACTIVITIES

1. Play G Major scale: G A B C D E F# G. Notice the F#.
2. Play D Major scale: D E F# G A B C# D. Notice two sharps.
3. Play F Major scale: F G A Bb C D E F. Notice the Bb.
4. Draw the Circle of Fifths from memory. Check your work with the table above.

Lesson 17 — Observe and Analyse: Major or Minor? (Hearing Method)

Introduction

One of the most amazing skills in music is being able to HEAR whether a song is major or minor — just by listening. This skill takes practice but it gets easy once your ear is trained.

Major vs Minor — The Emotional Difference

Major Scale	Minor Scale
Sounds happy, joyful, bright	Sounds sad, dark, mysterious
Like a sunny day	Like a rainy night
Used in: pop songs, anthems	Used in: sad ballads, dark themes
Example: 'Happy Birthday'	Example: 'Fur Elise'
Example: 'Twinkle Twinkle'	Example: 'Nothing Else Matters'

How to Identify Major or Minor by Ear

1. Listen to the song for 10-15 seconds.
2. Ask yourself: 'Does this make me feel happy or sad?'
3. Happy feeling = probably Major.
4. Sad, emotional, mysterious feeling = probably Minor.
5. Now test it: play the Major scale of the key, then the Minor. Which one matches the song?

Happy/Bright = MAJOR. Sad/Dark = MINOR. (This works 80% of the time for beginners!)

🎵 **Practice Song Examples**

MAJOR: 'Can't Stop the Feeling' (Justin Timberlake) — bright and danceyMAJOR: 'Walking on Sunshine' — energeticMINOR: 'Mad World' (Gary Jules) — sad, reflectiveMINOR: 'Stairway to Heaven' (beginning) — mysteriousMINOR: 'Hotel California' — haunting

The Shortcut — Listen for the 3rd Note

The key difference between major and minor is the 3rd note of the scale. If the melody uses the LOWER 3rd (closer to root), it sounds minor. The HIGHER 3rd sounds major.

C Major 3rd = E (4 semitones from C). C Minor 3rd = Eb (3 semitones from C).

✏️ PRACTICE ACTIVITIES

1. Listen to 10 songs. Write each as Major or Minor. Then check online if you were right.
2. Play C Major chord (C-E-G). Now play C Minor chord (C-Eb-G). Hear the difference.
3. Listen to 'Mad World' and 'Happy' (Pharrell). Which is Major? Which is Minor?
4. Ask your teacher to play random chords — identify Major or Minor without looking.

Lesson 18 — Seeing Scales Visually on the Octaves of the Keyboard

The Keyboard Layout

A standard keyboard has 88 keys — 52 white keys and 36 black keys. The notes repeat in a pattern every 12 keys. Each group of 12 keys is called an OCTAVE. There are about 7 full octaves on a standard piano.

One Octave — The Pattern

Look at the pattern of black keys: they appear in groups of 2 and 3. This pattern helps you find any note quickly:

- C is always just to the LEFT of the group of 2 black keys.
- F is always just to the LEFT of the group of 3 black keys.

Find C → Count up: C D E F G A B → then C again (next octave)

Seeing C Major Scale on the Keyboard

C Major uses ONLY white keys: C D E F G A B C. On the keyboard, it is a straight line of 8 white keys starting from C.

Visual Tip
Put a sticker on every C key (just left of 2 black keys). Now you can see all C notes at a glance. Each scale always starts and ends on the same letter but in a higher or lower octave.

Seeing Minor Scales — Black Keys Appear

Minor scales and scales other than C Major will use some black keys. For example:

- A Minor: A B C D E F G A — all white keys (same as C Major, starts on A)
- G Major: G A B C D E F# G — F# is a black key (one black key)
- D Major: D E F# G A B C# D — two black keys

🎵 **Visual Exercise**
Look at your keyboard. Starting from G:G (white) - A (white) - B (white) - C (white) - D (white) - E (white) - F# (BLACK) - G (white)You can SEE the scale going up with one black key.

✏️ PRACTICE ACTIVITIES

1. Place your index finger on every C key on the keyboard. Count how many there are.
2. Play C Major scale with eyes open. Then close your eyes and find every note by feel.
3. Play G Major scale and visually locate where the black key (F#) appears.
4. Mark all the notes of D Major on a paper keyboard diagram with coloured dots.

Lesson 19 — Same Tune but in Different Scales (Transposition)

What is Transposition?

Transposition means playing the same tune but starting on a different note — which means it will be in a different key. The SHAPE of the melody stays the same, but all the notes move up or down by the same amount.

Why Do We Transpose?

- A singer's voice might be too high or too low — transposing makes it comfortable.
- Different keys have different feelings, even for the same tune.
- Some keys are easier to play on guitar (open chords), others on piano.

Same INTERVAL pattern + Different starting note = Same tune, different key

How to Transpose — Step by Step

1. Learn the tune in the original key (e.g., C Major: C C G G A A G).
2. Identify the intervals between notes (e.g., same-same-up5-same-up2-same-down2).
3. Choose a new starting note (e.g., G).
4. Apply the same intervals from the new starting note.
5. In G Major: G G D D E E D — same shape, different key!

🎵 **Transposing Twinkle Twinkle**

In C Major: C C G G A A G F F E E D D CIn G Major: G G D D E E D C C B B A A GIn F Major: F F C C D D C Bb Bb A A G G FSame tune — but shifted to a new home note each time.

Key	First 7 Notes of Twinkle Twinkle
C Major	C C G G A A G
G Major	G G D D E E D
F Major	F F C C D D C
D Major	D D A A B B A

✏️ PRACTICE ACTIVITIES

1. Play Twinkle Twinkle in C Major. Then transpose it to G Major.
2. Play Happy Birthday starting on C. Then start on G. Then on F.
3. Pick a simple 4-note melody. Transpose it into 3 different keys.
4. Listen to two versions of the same song in different keys. Notice how the feel changes.

Lesson 20 — Listen and Follow the Shift of Scales (Modulation)

What is Modulation?

Modulation is when a song changes from one key (scale) to another KEY within the same song. It is different from transposition — in modulation, the song shifts key while playing, not from the beginning.

Why Do Songs Modulate?

- To create excitement or emotional uplift.
- To give the chorus a stronger, lifted feeling.
- To keep the song interesting and prevent boredom.

Types of Modulation

Type	Description
Direct (Abrupt)	Sudden jump to new key with no preparation. Used for surprise.
Pivot Chord	A chord that belongs to BOTH the old key and new key is used to transition smoothly.
Up a Half Step	Song goes up by one semitone (very common in pop: from C to C#).
Up a Whole Step	Song goes up by two semitones (e.g., from C to D).

Modulation = Current Key → Bridge/Pivot → New Key (usually up a step)

🎵 **Famous Modulation Examples**

'I Will Always Love You' (Whitney Houston) — the final chorus modulates UP by half a step, creating a soaring emotional lift.'Man in the Mirror' (Michael Jackson) — key shift near the final chorus.'Livin' on a Prayer' (Bon Jovi) — famous key change going into the last chorus.'Happy Birthday' — often sung in one key, then the last verse goes up.

How to Hear a Modulation

1. Listen carefully for a point in the song where everything suddenly sounds higher or lower.
2. Often accompanied by a pause, build-up, or a big emotional moment.
3. After the shift, try to find the new root note on your keyboard.

✏️ PRACTICE ACTIVITIES

1. Listen to 'I Will Always Love You'. Find the point where the key changes. What beat does it happen on?
2. Play Twinkle Twinkle in C Major for 4 bars, then suddenly switch to D Major for 4 bars.
3. Practice moving from C Major chord to Db Major chord smoothly.
4. Find 3 songs on YouTube that have a key change. Note the time (e.g., 2:30) where the change happens.

PART 4 — CHORDS

Chords are the foundation that supports the melody.

Lesson 21 — Write and Play the Primary Chords

What is a Chord?

A chord is three or more notes played at the same time. The combination of notes creates a harmony that supports the melody.

What are Primary Chords?

Primary chords are the three MOST IMPORTANT chords in any key. They are built on the 1st, 4th, and 5th notes of the scale. They are sometimes called the I, IV, and V chords.

Primary Chords = Chord I (Tonic) + Chord IV (Subdominant) + Chord V (Dominant)

Primary Chords in C Major

Chord	Notes
I — C Major (Tonic)	C - E - G
IV — F Major (Subdominant)	F - A - C
V — G Major (Dominant)	G - B - D

How to Build a Major Chord

Every major chord follows this formula:

Major Chord = Root + Major 3rd (4 semitones up) + Perfect 5th (7 semitones up)

Example: C Major = C + E (4 semitones up from C) + G (7 semitones up from C)

How to Write Chord Notation

- C = C Major chord
- Cm = C Minor chord
- F = F Major chord
- G = G Major chord

🎵 **Most Pop Songs Use Only I-IV-V**

C - F - G - C'La Bamba', 'Twist and Shout', 'Wild Thing', '12-bar blues', 'Louie Louie'All these famous songs use mainly just I, IV, and V chords!

Guitar Chord Shapes — C, F, G

On guitar, learn these open chord shapes:

- C Major: Fret positions (x32010) — thumb/basic form
- F Major: Barre chord at 1st fret (133211) — or simplified F (xx3211)
- G Major: (320003) — one of the most used guitar chords

✏️ **PRACTICE ACTIVITIES**

Play the I-IV-V progression in C Major: C - F - G - C. Repeat 10 times.

Build the I-IV-V chords in G Major (G-C-D). Write the notes for each chord.

Play the 12-bar blues pattern using only C, F, and G chords.

Listen to 'La Bamba' and identify the I, IV, V chords.

Lesson 22 — Write and Play Secondary Chords

What are Secondary Chords?

Secondary chords are chords built on the other scale degrees — the 2nd, 3rd, and 6th notes of the scale. They add colour and variety to music. Without them, songs would be repetitive and plain.

Secondary Chords in C Major

Chord	Notes
II — D Minor (Supertonic)	D - F - A
III — E Minor (Mediant)	E - G - B
VI — A Minor (Submediant)	A - C - E
VII° — B Diminished (Leading)	B - D - F

Minor Chord Formula

Minor Chord = Root + Minor 3rd (3 semitones up) + Perfect 5th (7 semitones up)

Example: D Minor = D + F (3 semitones from D) + A (7 semitones from D)

The vi Chord — Most Emotional

The vi chord (6th degree minor) is extremely popular in pop and emotional music. In C Major, it is A Minor (Am). The famous '50s progression uses it heavily:

I - VI - IV - V = C - Am - F - G (In C Major)

🎵 **Songs Using I-vi-IV-V**

'Can't Help Falling in Love' (Elvis)'Let It Be' (Beatles)'No Woman No Cry' (Bob Marley)'Stand by Me''Every Breath You Take' (The Police)

✏️ **PRACTICE ACTIVITIES**

GRADE 2 : Listen and Play

1. Play the ii chord (Dm): D-F-A. Hold each note together on keyboard.
2. Play the vi chord (Am): A-C-E. Listen to its sad sound.
3. Play I-vi-IV-V: C Am F G. Repeat many times. This is THE most popular chord progression!
4. Find 2 songs that use Am chord. Listen and identify it.

Lesson 23 — Write and Play Overlapping Chords (Inversions)

What are Inversions?

An inversion is when you rearrange the notes of a chord so that a different note is on the bottom (the bass note). Instead of always having the root note at the bottom, you can put the 3rd or 5th at the bottom.

Types of Inversions

Inversion	Bottom Note
Root Position	Root note is at the bottom. Example: C-E-G
First Inversion	3rd is at the bottom. Example: E-G-C
Second Inversion	5th is at the bottom. Example: G-C-E

C Major Root: C-E-G | First Inversion: E-G-C | Second Inversion: G-C-E

Why Use Inversions?

- Smoother bass movement — instead of big jumps, the bass moves in small steps.
- Creates a flowing, connected sound between chords.
- Makes it easier to play — your hand does not have to jump far.

🎵 **Smooth Bass Line Using Inversions**

Instead of: C (root) → Am (root) = big jump from C to A in bassUse inversions: C (root: C-E-G) → Am/C (C in bass: C-A-E)The bass note stays on C! Very smooth and professional sound.

Slash Chord Notation

A slash chord like 'Am/C' means: play Am chord, but put C as the lowest note.

Chord / Bass Note = Slash Chord. Example: G/B = G Major chord with B in the bass

✏️ PRACTICE ACTIVITIES

1. Play C Major in all 3 positions: C-E-G, then E-G-C, then G-C-E.
2. Play F Major in first inversion: A-C-F.
3. Connect C → Am using inversions so the bass barely moves.
4. Learn the following sequence with inversions: C - G/B - Am - F (smooth bass: C-B-A-A).

Lesson 24 — Write and Play Seventh Chords

What is a Seventh Chord?

A seventh chord is a chord that has FOUR notes instead of three. You add one more note — the 7th note above the root — on top of the regular triad (3-note chord).

Seventh Chord = Root + 3rd + 5th + 7th

Types of Seventh Chords

Type	Formula & Example
Major 7th (Maj7)	Major triad + Major 7th. Cmaj7 = C-E-G-B. Warm, jazzy sound.
Dominant 7th (7)	Major triad + Minor 7th. C7 = C-E-G-Bb. Bluesy, tense sound.
Minor 7th (m7)	Minor triad + Minor 7th. Cm7 = C-Eb-G-Bb. Smooth, jazzy sound.
Half Diminished (m7b5)	Diminished triad + Minor 7th. Bm7b5 = B-D-F-A. Unstable sound.
Diminished 7th (dim7)	Diminished triad + Diminished 7th. Bdim7 = B-D-F-Ab. Very tense.

The Most Important: G7 (Dominant 7th)

The G7 chord (G-B-D-F) is the most commonly used 7th chord. It creates strong tension that WANTS to resolve back to C Major. This tension and resolution is the basis of all Western music harmony.

G7 → C (Tension → Resolution) = The most satisfying movement in Western music

🎵 **7th Chords in Famous Songs**

Cmaj7: 'Dreams' by Fleetwood Mac (opening)C7: Blues songs (every 12-bar blues uses 7th chords)Am7: 'Superstition' by Stevie WonderDm7 G7 Cmaj7: Classic jazz progression (ii-V-I)

The ii-V-I Jazz Progression

In C Major: Dm7 → G7 → Cmaj7. This is THE fundamental jazz progression. Almost every jazz standard uses it.

✏️ PRACTICE ACTIVITIES

1. Play Cmaj7: C-E-G-B. Hold all 4 notes. Listen to the warm, dreamy sound.
2. Play G7: G-B-D-F. Notice the tense feeling. Then resolve to C: G7 → C.
3. Play the ii-V-I: Dm7 → G7 → Cmaj7. Repeat 10 times.
4. Build all 7th chords on each note of C Major scale. Write them all out.

Lesson 25 — Chord Progressions

What is a Chord Progression?

A chord progression is a series of chords played in a specific order that creates a harmonic pattern. Chord progressions are the backbone of almost every song you hear. They create the emotional journey of the music.

How Chord Progressions Are Written

Chord progressions are written using Roman numerals to show the position in the scale:

- I = Tonic (home, stable)
- ii = Supertonic (minor)
- iii = Mediant (minor)
- IV = Subdominant (warm, moving away from home)
- V = Dominant (tension, wants to go home to I)
- vi = Submediant (sad, emotional)
- vii° = Leading tone (very tense)

The Most Famous Chord Progressions in Music

Progression Name	Roman Numerals & C Major Notes
I-IV-V-I (Basic)	C - F - G - C. Foundation of rock, blues, country.
I-V-vi-IV (Pop)	C - G - Am - F. THE modern pop progression.
I-vi-IV-V (50s)	C - Am - F - G. Doo-wop, classic pop.
ii-V-I (Jazz)	Dm - G7 - C. Jazz standard.
I-IV-I-V (Blues)	C - F - C - G. 12-bar blues base.
vi-IV-I-V (Minor feel)	Am - F - C - G. Emotional, modern pop.

The I-V-vi-IV — Pop Music's Most Used Progression

In C Major: C - G - Am - F

This 4-chord progression is used in THOUSANDS of pop songs. Once you learn it, you can play along with most popular songs.

🎵 **Songs Using C-G-Am-F (I-V-vi-IV)**

'Let Her Go' (Passenger)'Someone Like You' (Adele — Am-F-C-G)'With or Without You' (U2)'Wonderwall' (Oasis — Em7-G-Dsus4-A7)'Despacito"No Woman No Cry' (Bob Marley)

How to Build Your Own Progression

1. Start on the I chord (home base).
2. Choose 3 more chords from the scale (any of ii, iii, IV, V, vi).
3. End on V to create tension, or I to feel finished.
4. Try different orders. Listen. Trust your ears!

Great starting formula: I → vi → IV → V → I

The 12-Bar Blues

The 12-bar blues is one of the oldest and most important chord progressions in Western music. It uses only I, IV, and V chords:

Bar	Chord (in C)
Bar 1	C (I)
Bar 2	C (I)
Bar 3	C (I)
Bar 4	C (I)
Bar 5	F (IV)
Bar 6	F (IV)
Bar 7	C (I)
Bar 8	C (I)
Bar 9	G (V)
Bar 10	F (IV)
Bar 11	C (I)
Bar 12	G (V) — turns around to bar 1

✏️ PRACTICE ACTIVITIES

Play the I-V-vi-IV progression in C Major (C-G-Am-F). Repeat 20 times until smooth.

Play the 12-bar blues in C Major. Use C7, F7, G7 for a more authentic blues sound.

Put on any pop song and try to play along using only C-G-Am-F. Does it fit?

Compose your own chord progression using at least 4 chords. Write it down and play it every day for a week.

Play the same chord progression in 2 different keys (e.g., C Major and G Major).

Music Theory Formula Reference

Semitone-Based · Scale-Degree-Based · Both Methods Side by Side

SECTION A — Semitone-Based Formulas

Semitone formulas count the exact number of semitones (half-steps) between each note. This method works in all 12 keys without knowing any scale. Best for: keyboard players building chords by ear, quick transposition, and understanding interval distances.

A1 — Chord Semitone Formulas (all groups)

Group	Symbol	Chord Name	Semitone Formula (R = Root, + = semitones apart)	Step Count	Example (Key of C)	Sound / Feel
Triad	Maj	Major	R + 4 + 3	4 — 3 st	C — E — G	Bright, happy
	min	Minor	R + 3 + 4	3 — 4 st	C — Eb — G	Dark, emotional
	dim	Diminished	R + 3 + 3	3 — 3 st	C — Eb — Gb	Tense, unstable
	aug	Augmented	R + 4 + 4	4 — 4 st	C — E — G#	Eerie, unresolved
Sus	sus2	Suspended 2	R + 2 + 5	2 — 5 st	C — D — G	Open, floating
	sus4	Suspended 4	R + 5 + 2	5 — 2 st	C — F — G	Anticipation
7th	Maj7	Major 7th	R + 4 + 3 + 4	4 — 3 — 4 st	C — E — G — B	Bright, dreamy
	7	Dominant 7th	R + 4 + 3 + 3	4 — 3 — 3 st	C — E — G — Bb	Bluesy, tense
	m7	Minor 7th	R + 3 + 4 + 3	3 — 4 — 3 st	C — Eb — G — Bb	Smooth, soulful
	mMaj7	Minor Major 7th	R + 3 + 4 + 4	3 — 4 — 4 st	C — Eb — G — B	Dark, cinematic
	m7b5	Half Diminished	R + 3 + 3 + 4	3 — 3 — 4 st	C — Eb — Gb — Bb	Tense, jazz ii
	dim7	Diminished 7th	R + 3 + 3 + 3	3 — 3 — 3 st	C — Eb — Gb — A	Fully tense, dramatic
7sus	7sus2	7 Suspended 2	R + 2 + 5 + 3	2 — 5 — 3 st	C — D — G — Bb	Lush, worship
	7sus4	7 Suspended 4	R + 5 + 2 + 3	5 — 2 — 3 st	C — F — G — Bb	Floaty, unresolved

A2 — Scale Semitone Formulas

Scale	Step Pattern (W=2 st, H=1 st)	Semitone Steps from Root	Sound / Feel
Major	W — W — H — W — W — W — H	2 — 2 — 1 — 2 — 2 — 2 — 1 semitones	Bright, resolved, uplifting
Natural Minor	W — H — W — W — H — W — W	2 — 1 — 2 — 2 — 1 — 2 — 2 semitones	Dark, emotional, expressive

Scale	Step Pattern (W=2 st, H=1 st)	Semitone Steps from Root	Sound / Feel
Harmonic Minor	W — H — W — W — H — +3 — H	2 — 1 — 2 — 2 — 1 — 3 — 1 semitones (+3 = Aug 2nd)	Exotic, tense, Middle-Eastern flavour

Major Scale — Degree & Semitone Breakdown:

Root	2nd	3rd	4th	5th	6th	7th	Oct
—	W (+2)	W (+2)	H (+1)	W (+2)	W (+2)	W (+2)	H (+1)
0 st	2 st	4 st	5 st	7 st	9 st	11 st	12 st

Natural Minor Scale — Degree & Semitone Breakdown:

Root	2nd	b3rd	4th	5th	b6th	b7th	Oct
—	W (+2)	H (+1)	W (+2)	W (+2)	H (+1)	W (+2)	W (+2)
0 st	2 st	3 st	5 st	7 st	8 st	10 st	12 st

Harmonic Minor Scale — Degree & Semitone Breakdown:

Root	2nd	b3rd	4th	5th	b6th	Maj 7th	Oct
—	W (+2)	H (+1)	W (+2)	W (+2)	H (+1)	+3 (Aug 2nd)	H (+1)
0 st	2 st	3 st	5 st	7 st	8 st	11 st	12 st

SECTION B — Scale-Degree Formulas

Scale-degree formulas name each note by its position number in the major scale (1 through 7), with "b" (flat) or "#" (sharp) to show alterations. This is the language of music theory, the Nashville Number System, and Roman numeral analysis. Best for: understanding chord function, analysing progressions, and communicating with other musicians.

B1 — Chord Scale-Degree Formulas (all groups)

Group	Symbol	Chord Name	Scale Degree Formula (b = flat, # = sharp)	Built From	Example (Key of C)	Sound / Feel
Triad	Maj	Major	1 — 3 — 5	Major scale	C — E — G	Bright, happy
	min	Minor	1 — b3 — 5	Major scale (b3)	C — Eb — G	Dark, emotional
	dim	Diminished	1 — b3 — b5	Major scale (b3, b5)	C — Eb — Gb	Tense, unstable
	aug	Augmented	1 — 3 — #5	Major scale (#5)	C — E — G#	Eerie, unresolved
Sus	sus2	Suspended 2	1 — 2 — 5	Major scale (no 3rd)	C — D — G	Open, floating
	sus4	Suspended 4	1 — 4 — 5	Major scale (no 3rd)	C — F — G	Anticipation
7th	Maj7	Major 7th	1 — 3 — 5 — 7	Major scale	C — E — G — B	Bright, dreamy

Group	Symbol	Chord Name	Scale Degree Formula (b = flat, # = sharp)	Built From	Example (Key of C)	Sound / Feel
	7	Dominant 7th	1 — 3 — 5 — b7	Mixolydian / Major V	C — E — G — Bb	Bluesy, tense
	m7	Minor 7th	1 — b3 — 5 — b7	Natural minor scale	C — Eb — G — Bb	Smooth, soulful
	mMaj7	Minor Major 7th	1 — b3 — 5 — 7	Harmonic minor scale	C — Eb — G — B	Dark, cinematic
	m7b5	Half Diminished	1 — b3 — b5 — b7	Locrian mode	C — Eb — Gb — Bb	Tense, jazz ii
	dim7	Diminished 7th	1 — b3 — b5 — bb7	Diminished scale	C — Eb — Gb — A	Fully tense, dramatic
7sus	7sus2	7 Suspended 2	1 — 2 — 5 — b7	Mixolydian (no 3rd)	C — D — G — Bb	Lush, worship
	7sus4	7 Suspended 4	1 — 4 — 5 — b7	Mixolydian (no 3rd)	C — F — G — Bb	Floaty, unresolved

B2 — Scale Degree-Number Formulas

Scale	Degree Numbers (b = flattened)	Notes in C	Sound / Feel
Major	1 — 2 — 3 — 4 — 5 — 6 — 7	C — D — E — F — G — A — B	Bright, resolved, uplifting
Natural Minor	1 — 2 — b3 — 4 — 5 — b6 — b7	C — D — Eb — F — G — Ab — Bb	Dark, emotional, expressive
Harmonic Minor	1 — 2 — b3 — 4 — 5 — b6 — 7	C — D — Eb — F — G — Ab — B	Exotic, tense, Middle-Eastern flavour

SECTION C — Both Methods Side by Side ✓

Most chords and scales can be expressed using BOTH semitone counting AND scale-degree numbers. The table below shows both formulas for every chord, with a note on when each method is more useful. "Both" means: use semitones when building/transposing; use degrees when analysing or communicating chord function.

C1 — Chords: Both Formulas + Usage Guide

Symbol	Chord Name	Semitone Formula →	Scale Degree Formula →	Example (C)	When to use which
Maj	Major	R + 4 + 3	1 — 3 — 5	C — E — G	Both equally useful — semitones for building, degrees for understanding function
min	Minor	R + 3 + 4	1 — b3 — 5	C — Eb — G	Both equally useful — semitones for building, degrees for understanding function

Symbol	Chord Name	Semitone Formula →	Scale Degree Formula →	Example (C)	When to use which
dim	Diminished	R + 3 + 3	1 — b3 — b5	C — Eb — Gb	Both equally useful — semitones for building, degrees for understanding function
aug	Augmented	R + 4 + 4	1 — 3 — #5	C — E — G#	Semitones: clearest — equal whole tones. Degrees (#5): shows raised 5th vs major
sus2	Suspended 2	R + 2 + 5	1 — 2 — 5	C — D — G	Degrees show missing 3rd clearly. Semitones useful for quick transposition
sus4	Suspended 4	R + 5 + 2	1 — 4 — 5	C — F — G	Degrees show missing 3rd clearly. Semitones useful for quick transposition
Maj7	Major 7th	R + 4 + 3 + 4	1 — 3 — 5 — 7	C — E — G — B	Both equally useful — semitones for building, degrees for understanding function
7	Dominant 7th	R + 4 + 3 + 3	1 — 3 — 5 — b7	C — E — G — Bb	Semitones: great for transposing quickly. Degrees: shows its role as V chord (tension→resolution)
m7	Minor 7th	R + 3 + 4 + 3	1 — b3 — 5 — b7	C — Eb — G — Bb	Both equally useful — semitones for building, degrees for understanding function
mMaj7	Minor Major 7th	R + 3 + 4 + 4	1 — b3 — 5 — 7	C — Eb — G — B	Semitones: easiest way to build it. Degrees: shows it comes from harmonic minor scale
m7b5	Half Diminished	R + 3 + 3 + 4	1 — b3 — b5 — b7	C — Eb — Gb — Bb	Semitones: easy to build by ear. Degrees (1-b3-b5-b7): shows it as flattened m7 chord
dim7	Diminished 7th	R + 3 + 3 + 3	1 — b3 — b5 — bb7	C — Eb — Gb — A	Semitones: shows equal-interval symmetry. Degrees: bb7 clarifies the double-flat 7th
7sus2	7 Suspended 2	R + 2 + 5 + 3	1 — 2 — 5 — b7	C — D — G — Bb	Degrees show missing 3rd clearly. Semitones useful for quick transposition
7sus4	7 Suspended 4	R + 5 + 2 + 3	1 — 4 — 5 — b7	C — F — G — Bb	Degrees show missing 3rd clearly. Semitones useful for quick transposition

C2 — Scales: Both Formulas

Scale	Semitone Steps →	Degree Numbers →	Notes in C	Notes
Major	W — W — H — W — W — W — H	1 — 2 — 3 — 4 — 5 — 6 — 7	C — D — E — F — G — A — B	W-W-H pattern is the same across all 12 keys — purely interval-based thinking
Natural Minor	W — H — W — W — H — W — W	1 — 2 — b3 — 4 — 5 — b6 — b7	C — D — Eb — F — G — Ab — Bb	Same notes as relative major — b3, b6, b7 show how it differs from major
Harmonic Minor	W — H — W — W — H — +3 — H	1 — 2 — b3 — 4 — 5 — b6 — 7	C — D — Eb — F — G — Ab — B	Only harmonic minor has an augmented 2nd (+3 st) step — neither W nor H alone

Major — Semitone grid (same as Section A):

Root	2nd	3rd	4th	5th	6th	7th	Oct
—	W (+2)	W (+2)	H (+1)	W (+2)	W (+2)	W (+2)	H (+1)
0 st	2 st	4 st	5 st	7 st	9 st	11 st	12 st

Natural Minor — Semitone grid (same as Section A):

Root	2nd	b3rd	4th	5th	b6th	b7th	Oct
—	W (+2)	H (+1)	W (+2)	W (+2)	H (+1)	W (+2)	W (+2)
0 st	2 st	3 st	5 st	7 st	8 st	10 st	12 st

Harmonic Minor — Semitone grid (same as Section A):

Root	2nd	b3rd	4th	5th	b6th	Maj 7th	Oct
—	W (+2)	H (+1)	W (+2)	W (+2)	H (+1)	+3 (Aug 2nd)	H (+1)
0 st	2 st	3 st	5 st	7 st	8 st	11 st	12 st

Key: R = Root · st = semitones · W = Whole step (2 st) · H = Half step (1 st) · b = flat (−1 st) · # = sharp (+1 st) · bb = double flat (−2 st) · P = Perfect · Maj = Major · Min = Minor · Dim = Diminished · Aug = Augmented

Music Theory Reference

Major & Minor Chords · Scales · Diatonic Charts · 7th Chords · Sus Chords

1. All 12 Major Chords

A major chord is a set of 3 notes — the Root, Major 3rd, and Perfect 5th — played together. It has a bright, happy, and confident sound. Every major chord follows the exact same formula regardless of the root note: move up 4 semitones from the root to get the Major 3rd, then 3 more semitones to get the Perfect 5th. The colour of the note tells you its role: green = root, amber = 3rd, blue = 5th.

Formula: Root + 4 semitones (Major 3rd) + 3 semitones (Perfect 5th)

Chord	Root	Major 3rd	Perfect 5th
C	C	E	G
Db	Db	F	Ab
D	D	F#	A
Eb	Eb	G	Bb
E	E	G#	B
F	F	A	C
F#	F#	A#	C#
G	G	B	D
Ab	Ab	C	Eb
A	A	C#	E
Bb	Bb	D	F
B	B	D#	F#

2. All 12 Minor Chords

A minor chord also has 3 notes — Root, Minor 3rd, and Perfect 5th — but the 3rd is one semitone lower than in a major chord. This small difference gives minor chords their dark, emotional, and serious sound. Formula: Root + 3 semitones = Minor 3rd, then + 4 semitones = Perfect 5th. Compared to major, the 3rd is flattened by 1 semitone. This is the most important chord family in minor key music and worship.

Formula: Root + 3 semitones (Minor 3rd) + 4 semitones (Perfect 5th)

Chord	Root	Minor 3rd	Perfect 5th
Am	A	C	E

Chord	Root	Minor 3rd	Perfect 5th
Bbm	**Bb**	Db	F
Bm	**B**	D	F#
Cm	**C**	Eb	G
C#m	**C#**	E	G#
Dm	**D**	F	A
Ebm	**Eb**	Gb	Bb
Em	**E**	G	B
Fm	**F**	Ab	C
F#m	**F#**	A	C#
Gm	**G**	Bb	D
G#m	**G#**	B	D#

3. Major Scales (All 12 Keys)

A scale is a set of notes arranged in a specific step pattern. The major scale uses 7 notes and always follows the pattern W W H W W W H (Whole-Whole-Half-Whole-Whole-Whole-Half). It has a bright, resolved, and uplifting sound. All major chords, diatonic progressions, and key signatures are built directly from this scale. Every key simply starts this same pattern from a different root note.

Pattern: W W H W W W H (W = whole step = 2 semitones, H = half step = 1 semitone)

Key	Scale Notes (W W H W W W H)
C	**C D E F G A B**
G	**G A B C D E F#**
D	**D E F# G A B C#**
A	**A B C# D E F# G#**
E	**E F# G# A B C# D#**
B	**B C# D# E F# G# A#**
F#	**F# G# A# B C# D# E#**
Db	**Db Eb F Gb Ab Bb C**
Ab	**Ab Bb C Db Eb F G**
Eb	**Eb F G Ab Bb C D**

Key	Scale Notes (W W H W W W H)
Bb	Bb C D Eb F G A
F	F G A Bb C D E

4. Natural Minor Scales (All 12 Keys)

The natural minor scale (also called the Aeolian mode) uses the pattern W H W W H W W. It has 7 notes just like the major scale but sounds darker and more emotional. Every minor scale is actually the same notes as its relative major — just starting from a different point (the 6th degree). This scale is the backbone of minor key songs, worship music in minor, rock, and most emotional film scores.

Pattern: W H W W H W W

Key	Scale Notes (W H W W H W W)
Am	A B C D E F G
Em	E F# G A B C D
Bm	B C# D E F# G A
F#m	F# G# A B C# D E
C#m	C# D# E F# G# A B
G#m	G# A# B C# D# E F#
Ebm	Eb F Gb Ab Bb B Db
Bbm	Bb C Db Eb F Gb Ab
Fm	F G Ab Bb C Db Eb
Cm	C D Eb F G Ab Bb
Gm	G A Bb C D Eb F
Dm	D E F G A Bb C

5. Major Keys — Diatonic Chords (Primary: I, IV, V)

Diatonic chords are all the chords you can build using only the notes from a given scale. In any major key there are 7 diatonic chords, one built on each scale degree. The Roman numerals tell you which degree of the scale the chord is built on. Uppercase = major chord, lowercase = minor chord, ° = diminished. The PRIMARY chords (I, IV, V) are the most commonly used in songs — they form the foundation of almost every popular chord progression in Western music.

Green = Primary chords (I, IV, V). Uppercase = major chord, lowercase = minor chord.

Key	I	ii	iii	IV	V	vi	vii°
C	C	Dm	Em	F	G	Am	Bdim
G	G	Am	Bm	C	D	Em	F#dim
D	D	Em	F#m	G	A	Bm	C#dim
A	A	Bm	C#m	D	E	F#m	G#dim
E	E	F#m	G#m	A	B	C#m	D#dim
B	B	C#m	D#m	E	F#	G#m	A#dim
F#	F#	G#m	A#m	B	C#	D#m	E#dim
Db	Db	Ebm	Fm	Gb	Ab	Bbm	Cdim
Ab	Ab	Bbm	Cm	Db	Eb	Fm	Gdim
Eb	Eb	Fm	Gm	Ab	Bb	Cm	Ddim
Bb	Bb	Cm	Dm	Eb	F	Gm	Adim
F	F	Gm	Am	Bb	C	Dm	Edim

6. Minor Keys — Diatonic Chords (Primary: i, VI, VII)

Just like major keys, minor keys have 7 diatonic chords built from the natural minor scale. The PRIMARY chords for minor are i, VI, and VII — these three chords power most minor key songs, worship music, film scores, and rock anthems. Notice that VI and VII are both MAJOR chords even though we are in a minor key, which gives that powerful, epic sound (e.g. Am - F - G in A minor). Chords are ordered by Roman numeral: i ii° III iv V VI VII.

Green = Primary chords (i, VI, VII) — natural minor / worship pattern. Order: i ii° III iv V VI VII

Key	i	ii°	III	iv	V	VI	VII
Am	Am	Bdim	C	Dm	E	F	G
Em	Em	F#dim	G	Am	B	C	D
Bm	Bm	C#dim	D	Em	F#	G	A
F#m	F#m	G#dim	A	Bm	C#	D	E
C#m	C#m	D#dim	E	F#m	G#	A	B
G#m	G#m	A#dim	B	C#m	D#	E	F#
Ebm	Ebm	Fdim	Gb	Abm	Bb	B	Db
Bbm	Bbm	Cdim	Db	Ebm	F	Gb	Ab
Fm	Fm	Gdim	Ab	Bbm	C	Db	Eb
Cm	Cm	Ddim	Eb	Fm	G	Ab	Bb

Key	i	ii°	III	iv	V	VI	VII
Gm	Gm	Adim	Bb	Cm	D	Eb	F
Dm	Dm	Edim	F	Gm	A	Bb	C

7. Relative Minor — All 12 Major Keys

Every major key has a relative minor that shares the exact same notes and key signature. The relative minor starts on the 6th degree (scale degree 6) of the major scale.

This is one of the most practical concepts in music. Because they share all the same notes, you can freely switch between a major key and its relative minor without changing any notes — only the tonal centre shifts. For example, C major and A minor share the exact same 7 notes (C D E F G A B). When the music feels bright and resolved, you are hearing it as C major. When it feels dark and emotional, you are hearing A minor. This is why so many songs use both in the same progression — for example Cmaj → Am or G → Em.

Quick rule: Major key root → go DOWN 3 semitones (or UP 9 semitones) = Relative minor root.

Major Key	Rel. Minor	6th Degree	Shared Notes (same key sig)	Key Signature
C	Am	A	C D E F G A B	0 sharps / flats
G	Em	E	G A B C D E F#	1 sharp (F#)
D	Bm	B	D E F# G A B C#	2 sharps
A	F#m	F#	A B C# D E F# G#	3 sharps
E	C#m	C#	E F# G# A B C# D#	4 sharps
B	G#m	G#	B C# D# E F# G# A#	5 sharps
F#	D#m	D#	F# G# A# B C# D# E#	6 sharps
Db	Bbm	Bb	Db Eb F Gb Ab Bb C	5 flats
Ab	Fm	F	Ab Bb C Db Eb F G	4 flats
Eb	Cm	C	Eb F G Ab Bb C D	3 flats
Bb	Gm	G	Bb C D Eb F G A	2 flats
F	Dm	D	F G A Bb C D E	1 flat (Bb)

8. Major Chord Inversions — Keyboard (Piano)

An inversion means rearranging the notes of a chord so that a note other than the root is at the bottom (bass). Every triad has 3 positions: Root Position (root in the bass), 1st Inversion (3rd in the bass), and 2nd Inversion (5th in the bass). On the keyboard, inversions let you keep your hand in one place while the harmony changes — this is called voice leading. The notes are always the same; only the lowest note changes.

Root Position = 1-3-5 | 1st Inversion = 3-5-1 | 2nd Inversion = 5-1-3 (numbers = scale degrees, read left to right = low to high)

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	Slash Note — 1st / 2nd
C	C — E — G	E — G — C	G — C — E	E / G
Db	Db — F — Ab	F — Ab — Db	Ab — Db — F	F / Ab
D	D — F# — A	F# — A — D	A — D — F#	F# / A
Eb	Eb — G — Bb	G — Bb — Eb	Bb — Eb — G	G / Bb
E	E — G# — B	G# — B — E	B — E — G#	G# / B
F	F — A — C	A — C — F	C — F — A	A / C
F#	F# — A# — C#	A# — C# — F#	C# — F# — A#	A# / C#
G	G — B — D	B — D — G	D — G — B	B / D
Ab	Ab — C — Eb	C — Eb — Ab	Eb — Ab — C	C / Eb
A	A — C# — E	C# — E — A	E — A — C#	C# / E
Bb	Bb — D — F	D — F — Bb	F — Bb — D	D / F
B	B — D# — F#	D# — F# — B	F# — B — D#	D# / F#

9. Minor Chord Inversions — Keyboard (Piano)

Minor Chord Inversions — same concept, same positions, just using the minor 3rd instead of major 3rd.

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	Slash Note — 1st / 2nd
Am	A — C — E	C — E — A	E — A — C	C / E
Bbm	Bb — Db — F	Db — F — Bb	F — Bb — Db	Db / F
Bm	B — D — F#	D — F# — B	F# — B — D	D / F#
Cm	C — Eb — G	Eb — G — C	G — C — Eb	Eb / G
C#m	C# — E — G#	E — G# — C#	G# — C# — E	E / G#
Dm	D — F — A	F — A — D	A — D — F	F / A
Ebm	Eb — Gb — Bb	Gb — Bb — Eb	Bb — Eb — Gb	Gb / Bb
Em	E — G — B	G — B — E	B — E — G	G / B
Fm	F — Ab — C	Ab — C — F	C — F — Ab	Ab / C

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	Slash Note — 1st / 2nd
F#m	F# — A — C#	A — C# — F#	C# — F# — A	A / C#
Gm	G — Bb — D	Bb — D — G	D — G — Bb	Bb / D
G#m	G# — B — D#	B — D# — G#	D# — G# — B	B / D#

10. Major Chord Inversions — Guitar

On guitar, inversions are created by changing which note sits on the lowest string being played. Root position means the root note is the lowest. 1st inversion means the 3rd is the lowest note (written as a slash chord e.g. C/E). 2nd inversion means the 5th is the lowest (e.g. C/G). Guitar inversions are used to create smoother basslines, keep chord shapes nearby on the neck, and add variety to strumming patterns. Slash chord notation tells you: Chord / Bass note.

Slash chord notation: C/E = C major with E in the bass (1st inversion) | C/G = C major with G in the bass (2nd inversion)

How to read fret notation: Each number = which fret to press on that string. x = mute the string. 0 = play it open (no finger). Strings are always written left to right: String 6 (Low E) → String 5 (A) → String 4 (D) → String 3 (G) → String 2 (B) → String 1 (High e).

String No.	String Name	Open Note	Fret Reading	Role in Chord
6	Low E	E2	*x = mute | 0 = open | 1-22 = fret no.*	Lowest / thickest string — bass notes live here
5	A	A2	*x = mute | 0 = open | 1-22 = fret no.*	2nd string — often carries bass in open chords
4	D	D3	*x = mute | 0 = open | 1-22 = fret no.*	3rd string — mid-range, core of most chords
3	G	G3	*x = mute | 0 = open | 1-22 = fret no.*	4th string — middle of the chord
2	B	B3	*x = mute | 0 = open | 1-22 = fret no.*	5th string — upper harmony notes
1	High e	E4	*x = mute | 0 = open | 1-22 = fret no.*	Highest / thinnest string — melody & top notes

Reading Example — C major and its inversions:

How to read	Str 6	Str 5	Str 4	Str 3	Str 2	Str 1	What it means
C major	x	3	2	0	1	0	Mute low E | Fret 3 on A | Fret 2 on D | Open G | Fret 1 on B | Open e
C/E (1st inv)	0	3	2	0	1	0	Open low E (= E note, the 3rd of C) | Same shape — only the bass note changed!
C/G (2nd inv)	3	3	2	0	1	0	Fret 3 on low E (= G note, the 5th of C) | G is now the lowest note

Each column = one string. Numbers in the inversion tables below follow the same left-to-right = String 6 to String 1 order.

Major Guitar Inversions — all 12 major chords in root position, 1st inversion, and 2nd inversion:

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	Practical Tip
C	x-3-2-0-1-0	C/E → 0-3-2-0-1-0	C/G → 3-3-2-0-1-0	*Open C shape; add low E for 2nd inv*
D	x-x-0-2-3-2	D/F# → 2-x-0-2-3-2	D/A → x-0-0-2-3-2	*Open D; use A string for 2nd inv*
E	0-2-2-1-0-0	E/G# → 4-x-2-1-0-0	E/B → x-2-2-1-0-0	*Open E; 2nd inv uses A string*
F	1-3-3-2-1-1	F/A → x-0-3-2-1-1	F/C → x-3-3-2-1-1	*Barre F; A string = 1st inv bass*
G	3-2-0-0-0-3	G/B → x-2-0-0-0-3	G/D → x-x-0-0-0-3	*Open G; B string easiest for 1st inv*
A	x-0-2-2-2-0	A/C# → x-4-2-2-2-0	A/E → 0-0-2-2-2-0	*Open A; low E gives 2nd inv*
Bb	x-1-3-3-3-1	Bb/D → x-5-3-3-3-1	Bb/F → 1-1-3-3-3-1	*Barre Bb; add low F for 2nd inv*
B	x-2-4-4-4-2	B/D# → x-6-4-4-4-2	B/F# → 2-2-4-4-4-2	*Barre B; low F# for 2nd inv*
Db	x-4-6-6-6-4	Db/F → 1-4-3-1-2-1	Db/Ab→ 4-4-6-6-6-4	*Barre shape; Ab on low E*
Eb	x-6-8-8-8-6	Eb/G → 3-6-5-3-4-3	Eb/Bb→ 6-6-8-8-8-6	*High barre; Bb on low E*
Ab	4-6-6-5-4-4	Ab/C → x-3-5-5-4-4	Ab/Eb→ x-6-6-5-4-4	*Barre Ab shape on 4th fret*
F#	2-4-4-3-2-2	F#/A#→ x-0-4-3-2-2	F#/C#→ x-4-4-3-2-2	*Barre F# shape on 2nd fret*

11. Minor Chord Inversions — Guitar

Minor Guitar Inversions — same logic, the slash note shows which note of the minor chord sits lowest on the neck.

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	Practical Tip
Am	x-0-2-2-1-0	Am/C → x-3-2-2-1-0	Am/E → 0-0-2-2-1-0	*Open Am; low E = 2nd inv*
Dm	x-x-0-2-3-1	Dm/F → 1-x-0-2-3-1	Dm/A → x-0-0-2-3-1	*Open Dm; A string for 2nd inv*
Em	0-2-2-0-0-0	Em/G → 3-2-0-0-0-0	Em/B → x-2-2-0-0-0	*Open Em; 2nd inv use B string*
Fm	1-3-3-1-1-1	Fm/Ab → 4-3-1-1-1-1	Fm/C → x-3-3-1-1-1	*Barre Fm; Ab gives 1st inv*

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	Practical Tip
Gm	3-5-5-3-3-3	Gm/Bb → x-1-3-3-3-3	Gm/D → x-5-5-3-3-3	*Barre Gm; Bb string = 1st inv*
Bm	x-2-4-4-3-2	Bm/D → x-5-4-4-3-2	Bm/F# → 2-2-4-4-3-2	*Barre Bm; F# on low E = 2nd inv*
C#m	x-4-6-6-5-4	C#m/E → 0-4-6-6-5-4	C#m/G#→ 4-4-6-6-5-4	*Barre C#m; open E = 1st inv*
F#m	2-4-4-2-2-2	F#m/A → x-0-4-2-2-2	F#m/C#→ x-4-4-2-2-2	*Barre F#m shape on 2nd fret*
G#m	4-6-6-4-4-4	G#m/B → x-2-6-4-4-4	G#m/D#→ x-6-6-4-4-4	*Barre G#m shape on 4th fret*
Bbm	x-1-3-3-2-1	Bbm/Db→ x-4-3-3-2-1	Bbm/F → 1-1-3-3-2-1	*Barre Bbm; F on low E = 2nd inv*
Cm	x-3-5-5-4-3	Cm/Eb → x-6-5-5-4-3	Cm/G → 3-3-5-5-4-3	*Barre Cm; G on low E = 2nd inv*
Ebm	x-6-8-8-7-6	Ebm/Gb→ 2-6-8-8-7-6	Ebm/Bb→ x-1-3-3-2-1	*High barre; Gb on low E*

12. All 7th Chords — Formula Reference

A 7th chord adds a 4th note — the 7th interval — on top of a basic triad. The type of 7th interval (major or minor) and the quality of the triad underneath (major or minor) combine to create 6 distinct 7th chord types, each with its own sound and emotional feel. 7th chords are used everywhere in jazz, gospel, worship, soul, and R&B because they add richness, colour, and tension to a basic triad. The table below summarises all 6 types with their formula and character.

7th chords add a 4th note on top of the triad. The type of 7th interval changes the colour and feel.

Chord Type	Symbol	Semitone Formula	Sound / Feel
Major 7th	Maj7	Root + 4 + 3 + 4 semitones	Bright, dreamy, jazzy
Dominant 7th	7	Root + 4 + 3 + 3 semitones	Bluesy, tense, wants to resolve
Minor 7th	m7	Root + 3 + 4 + 3 semitones	Smooth, mellow, soulful
Minor Major 7th	mMaj7	Root + 3 + 4 + 4 semitones	Dark, mysterious, cinematic
Half Diminished	m7b5	Root + 3 + 3 + 4 semitones	Tense, unstable, jazz ii chord
Diminished 7th	dim7	Root + 3 + 3 + 3 semitones	Fully tense, symmetrical, dramatic

13. Major 7th (Maj7) — All 12 Keys

The Major 7th chord is a major triad with a major 7th interval added on top. It has a lush, dreamy, and sophisticated sound — think of it as a major chord with an extra layer of brightness. Widely used in jazz, contemporary worship, and R&B. Example: CMaj7 = C E G B. A common worship substitution is to play Maj7 instead of a plain major chord on the I or IV to add colour without changing the function.

Formula: Root + 4 + 3 + 4 semitones | Root, Major 3rd, Perfect 5th, Major 7th

Chord	Root	Maj 3rd	P 5th	Maj 7th
CMaj7	C	E	G	B
DbMaj7	Db	F	Ab	C
DMaj7	D	F#	A	C#
EbMaj7	Eb	G	Bb	D
EMaj7	E	G#	B	D#
FMaj7	F	A	C	E
GbMaj7	Gb	Bb	Db	F
GMaj7	G	B	D	F#
AbMaj7	Ab	C	Eb	G
AMaj7	A	C#	E	G#
BbMaj7	Bb	D	F	A
BMaj7	B	D#	F#	A#

14. Dominant 7th (7) — All 12 Keys

The Dominant 7th chord is a major triad with a minor 7th on top. It is the most harmonically tense chord in music — it strongly wants to resolve to the tonic (home chord). It naturally occurs on the V (5th degree) of any major or minor scale, which is why it is called the Dominant. Example: G7 = G B D F, and it strongly pulls to C major. Blues, jazz, gospel, and classical music all rely heavily on this chord for tension and release.

Formula: Root + 4 + 3 + 3 semitones | Root, Major 3rd, Perfect 5th, Minor 7th

Chord	Root	Maj 3rd	P 5th	Min 7th
C7	C	E	G	A#
Db7	Db	F	Ab	B
D7	D	F#	A	C
Eb7	Eb	G	Bb	Db
E7	E	G#	B	D

Chord	Root	Maj 3rd	P 5th	Min 7th
F7	F	A	C	D#
Gb7	Gb	Bb	Db	E
G7	G	B	D	F
Ab7	Ab	C	Eb	Gb
A7	A	C#	E	G
Bb7	Bb	D	F	Ab
B7	B	D#	F#	A

15. Minor 7th (m7) — All 12 Keys

The Minor 7th chord is a minor triad with a minor 7th interval on top. It has a smooth, mellow, and soulful character — less tense than a dominant 7th, more colourful than a plain minor chord. The m7 chord naturally occurs on the ii, iii, and vi degrees of a major scale. It is one of the most used chords in jazz, neo-soul, R&B, and modern worship. Example: Am7 = A C E G.

Formula: Root + 3 + 4 + 3 semitones | Root, Minor 3rd, Perfect 5th, Minor 7th

Chord	Root	Min 3rd	P 5th	Min 7th
Cm7	C	D#	G	A#
Dbm7	Db	E	Ab	B
Dm7	D	F	A	C
Ebm7	Eb	Gb	Bb	Db
Em7	E	G	B	D
Fm7	F	G#	C	D#
Gbm7	Gb	A	Db	E
Gm7	G	A#	D	F
Abm7	Ab	B	Eb	Gb
Am7	A	C	E	G
Bbm7	Bb	Db	F	Ab
Bm7	B	D	F#	A

16. Minor Major 7th (mMaj7) — All 12 Keys

The Minor Major 7th chord combines a minor triad with a major 7th interval on top — a striking and unusual pairing that creates a dark, mysterious, cinematic sound. It comes from the harmonic minor scale (where the 7th degree is raised by one semitone). You hear this chord in film scores, jazz, and any music that needs a haunting or dramatic colour. Example: AmMaj7 = A C E G#. Use it as the i chord for a more dramatic minor sound.

Formula: Root + 3 + 4 + 4 semitones | Root, Minor 3rd, Perfect 5th, Major 7th

Chord	Root	Min 3rd	P 5th	Maj 7th
CmMaj7	C	D#	G	B
DbmMaj7	Db	E	Ab	C
DmMaj7	D	F	A	C#
EbmMaj7	Eb	Gb	Bb	D
EmMaj7	E	G	B	D#
FmMaj7	F	G#	C	E
GbmMaj7	Gb	A	Db	F
GmMaj7	G	A#	D	F#
AbmMaj7	Ab	B	Eb	G
AmMaj7	A	C	E	G#
BbmMaj7	Bb	Db	F	A
BmMaj7	B	D	F#	A#

17. Half Diminished (m7b5) — All 12 Keys

The Half Diminished chord (also written as ø or m7b5) is a diminished triad with a minor 7th on top. It sounds tense and unstable but not as harsh as the fully diminished chord. In jazz it is the standard ii chord in a minor ii-V-i progression (e.g. Bm7b5 - E7 - Am). It naturally occurs on the vii° of a major scale and the ii° of a natural minor scale. Example: Bm7b5 = B D F A.

Formula: Root + 3 + 3 + 4 semitones | Root, Minor 3rd, Diminished 5th, Minor 7th

Chord	Root	Min 3rd	Dim 5th	Min 7th
Cm7b5	C	D#	F#	A#
Dbm7b5	Db	E	G	B
Dm7b5	D	F	G#	C

Chord	Root	Min 3rd	Dim 5th	Min 7th
Ebm7b5	Eb	Gb	A	Db
Em7b5	E	G	A#	D
Fm7b5	F	G#	B	D#
Gbm7b5	Gb	A	C	E
Gm7b5	G	A#	C#	F
Abm7b5	Ab	B	D	Gb
Am7b5	A	C	D#	G
Bbm7b5	Bb	Db	E	Ab
Bm7b5	B	D	F	A

18. Diminished 7th (dim7) — All 12 Keys

The Diminished 7th chord stacks 4 notes each exactly 3 semitones apart, making it fully symmetrical. This symmetry means it has no clear root and sounds equally tense from any angle. It creates maximum harmonic tension and is used as a dramatic passing chord, in horror film scores, and as a chromatic approach chord in gospel and jazz. Example: Bdim7 = B D F Ab. Because of its symmetry, Bdim7 = Ddim7 = Fdim7 = Abdim7 — they are all the same chord in different inversions.

Formula: Root + 3 + 3 + 3 semitones | Root, Minor 3rd, Diminished 5th, Diminished 7th (bb7)

Chord	Root	Min 3rd	Dim 5th	Dim 7th
Cdim7	C	D#	F#	A
Dbdim7	Db	E	G	Bb
Ddim7	D	F	G#	B
Ebdim7	Eb	Gb	A	C
Edim7	E	G	A#	C#
Fdim7	F	G#	B	D
Gbdim7	Gb	A	C	Eb
Gdim7	G	A#	C#	E
Abdim7	Ab	B	D	F
Adim7	A	C	D#	F#
Bbdim7	Bb	Db	E	G
Bdim7	B	D	F	G#

19. Sus Chords — Formula Reference

Sus (suspended) chords are built by replacing the 3rd of a chord with either the 2nd or the 4th scale degree. Because the 3rd is removed, sus chords have no major or minor quality — they sound open, ambiguous, and floating. They are extremely popular in modern worship music, ambient, and cinematic music because they create tension without the harshness of a dissonant chord. Sus chords often resolve back to the major or minor version of the same chord (e.g. Gsus4 → G).

Sus chords replace the 3rd with the 2nd or 4th degree. No major or minor quality — open and floating.

Chord Type	Symbol	Semitone Formula	Sound / Feel
Sus2	sus2	Root + 2 + 5 semitones	Open, floating, unresolved
Sus4	sus4	Root + 5 + 2 semitones	Suspended, anticipating
7sus2	7sus2	Root + 2 + 5 + 3 semitones	Airy, modern worship sound
7sus4	7sus4	Root + 5 + 2 + 3 semitones	Classic suspended dominant

20. Sus2 — All 12 Keys

The Sus2 chord replaces the 3rd with the major 2nd (one whole step above the root). It sounds spacious, airy, and unresolved — like the chord is floating in mid-air. Sus2 chords are very popular in worship music because they avoid the definitive major/minor sound and work over multiple chord changes. Example: Csus2 = C D G. The missing 3rd is what gives it that open, modern sound heard in songs like What a Beautiful Name.

Formula: Root + 2 + 5 semitones | Root, Major 2nd, Perfect 5th (3rd omitted)

Chord	Root	Maj 2nd	P 5th
Csus2	C	D	G
Dbsus2	Db	Eb	Ab
Dsus2	D	E	A
Ebsus2	Eb	F	Bb
Esus2	E	F#	B
Fsus2	F	G	C
Gbsus2	Gb	Ab	Db
Gsus2	G	A	D
Absus2	Ab	Bb	Eb
Asus2	A	B	E
Bbsus2	Bb	C	F
Bsus2	B	C#	F#

21. Sus4 — All 12 Keys

The Sus4 chord replaces the 3rd with the perfect 4th. It creates a feeling of anticipation and tension that strongly wants to resolve back down to the major chord. Sus4 is one of the oldest and most recognisable harmonic devices in Western music — heard in classical cadences, rock power ballads, and modern pop. Example: Gsus4 = G C D, which naturally resolves to G major. The 4th degree feels like it is leaning toward the 3rd.

Formula: Root + 5 + 2 semitones | Root, Perfect 4th, Perfect 5th (3rd omitted)

Chord	Root	P 4th	P 5th
Csus4	C	F	G
Dbsus4	Db	Gb	Ab
Dsus4	D	G	A
Ebsus4	Eb	Ab	Bb
Esus4	E	A	B
Fsus4	F	A#	C
Gbsus4	Gb	B	Db
Gsus4	G	C	D
Absus4	Ab	Db	Eb
Asus4	A	D	E
Bbsus4	Bb	Eb	F
Bsus4	B	E	F#

22. 7sus2 — All 12 Keys

The 7sus2 chord combines the open sound of a sus2 with the warmth of a minor 7th. It has 4 notes: Root, Major 2nd, Perfect 5th, and Minor 7th. This chord is extremely popular in contemporary worship, Hillsong-style music, and neo-soul because it sounds lush and harmonic without committing to a major or minor quality. Example: Gsus2/7 (also written G7sus2) = G A D F. You can hold this chord over an entire bar while other chords move in the bass.

Formula: Root + 2 + 5 + 3 semitones | Root, Major 2nd, Perfect 5th, Minor 7th — popular in worship/neo-soul

Chord	Root	Maj 2nd	P 5th	Min 7th
C7sus2	C	D	G	A#
Db7sus2	Db	Eb	Ab	B
D7sus2	D	E	A	C
Eb7sus2	Eb	F	Bb	Db
E7sus2	E	F#	B	D

Chord	Root	Maj 2nd	P 5th	Min 7th
F7sus2	F	G	C	D#
Gb7sus2	Gb	Ab	Db	E
G7sus2	G	A	D	F
Ab7sus2	Ab	Bb	Eb	Gb
A7sus2	A	B	E	G
Bb7sus2	Bb	C	F	Ab
B7sus2	B	C#	F#	A

23. 7sus4 — All 12 Keys

The 7sus4 chord is a sus4 chord with a minor 7th added. It has 4 notes: Root, Perfect 4th, Perfect 5th, and Minor 7th. This is the classic suspended dominant chord — it sounds like a dominant 7th chord but with the 3rd replaced by the 4th, making it less harsh and more floaty. It is widely used in jazz (as a modern V chord) and gospel. Example: G7sus4 = G C D F. It can resolve to G7 (by dropping the 4th to the 3rd) and then to C major.

Formula: Root + 5 + 2 + 3 semitones | Root, Perfect 4th, Perfect 5th, Minor 7th

Chord	Root	P 4th	P 5th	Min 7th
C7sus4	C	F	G	A#
Db7sus4	Db	Gb	Ab	B
D7sus4	D	G	A	C
Eb7sus4	Eb	Ab	Bb	Db
E7sus4	E	A	B	D
F7sus4	F	A#	C	D#
Gb7sus4	Gb	B	Db	E
G7sus4	G	C	D	F
Ab7sus4	Ab	Db	Eb	Gb
A7sus4	A	D	E	G
Bb7sus4	Bb	Eb	F	Ab
B7sus4	B	E	F#	A

Maj7=R+4+3+4 | 7=R+4+3+3 | m7=R+3+4+3 | mMaj7=R+3+4+4 | m7b5=R+3+3+4 | dim7=R+3+3+3 | sus2=R+2+5 | sus4=R+5+2 (semitones)

24. Major 7th (Maj7) — Guitar Voicings

The Major 7th chord on guitar has that iconic lush, dreamy sound perfect for jazz, worship, and R&B. Because the guitar has 6 strings you can voice these chords in many ways — the shapes below are the most practical and commonly used positions. The Maj7 sound comes from keeping that bright major 7th note ringing above the triad.

Formula: Root + 4 + 3 + 4 semitones | Root, Maj 3rd, P 5th, Maj 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
CMaj7	x-3-2-0-0-0	C E G B	*Open shape — omit 5th from open C*
DbMaj7	x-4-3-1-1-1	Db F Ab C	*Barre on 1st fret, Db shape*
DMaj7	x-x-0-2-2-2	D F# A C#	*Open D shape — easy voicing*
EbMaj7	x-6-5-3-3-3	Eb G Bb D	*Barre 3rd fret — Db shape up 2*
EMaj7	0-2-2-1-0-0	E G# B D#	*Open E — lift 3rd finger from E*
FMaj7	1-3-3-2-1-0	F A C E	*Partial barre; open high e*
GbMaj7	2-4-4-3-2-1	Gb Bb Db F	*Full barre 2nd fret — F shape*
GMaj7	3-2-0-0-0-2	G B D F#	*Open G with F# on high e*
AbMaj7	4-6-5-5-4-4	Ab C Eb G	*Barre 4th fret — E shape*
AMaj7	x-0-2-1-2-0	A C# E G#	*Open — one of guitar's nicest chords*
BbMaj7	x-1-3-2-3-1	Bb D F A	*Partial barre; elegant voicing*
BMaj7	x-2-4-3-4-2	B D# F# A#	*Barre 2nd fret — A shape*

25. Dominant 7th (7) — Guitar Voicings

The Dominant 7th chord is the backbone of blues, jazz, and gospel on guitar. Its tension-and-release quality comes from the tritone between the major 3rd and minor 7th. Open shapes like E7, A7, D7, G7, and B7 are among the most-played chords in the guitar repertoire. For barre chord shapes, the E7 and A7 moveable forms cover all 12 keys.

Formula: Root + 4 + 3 + 3 semitones | Root, Maj 3rd, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
C7	x-3-2-3-1-0	C E G Bb	*Open shape — classic blues C7*
Db7	x-4-3-4-2-1	Db F Ab B	*Barre 1st — E7 shape up 1*
D7	x-x-0-2-1-2	D F# A C	*Open D7 — very common*

Eb7	**x-6-5-6-4-3**	**Eb G Bb Db**	*Barre 3rd — E7 shape up 3*
E7	**0-2-0-1-0-0**	**E G# B D**	*Open E7 — lift middle finger*
F7	**1-3-1-2-1-1**	**F A C Eb**	*Barre 1st — E7 shape*
Gb7	**2-4-2-3-2-2**	**Gb Bb Db E**	*Barre 2nd — E7 shape*
G7	**3-2-0-0-0-1**	**G B D F**	*Open G7 — pinky on high e*
Ab7	**4-6-4-5-4-4**	**Ab C Eb Gb**	*Barre 4th — E7 shape*
A7	**x-0-2-0-2-0**	**A C# E G**	*Open A7 — very common*
Bb7	**x-1-3-1-3-1**	**Bb D F Ab**	*Barre 1st — A7 shape*
B7	**x-2-1-2-0-2**	**B D# F# A**	*Open B7 — common in E major songs*

26. Minor 7th (m7) — Guitar Voicings

Minor 7th chords on guitar have a smooth, soulful quality — less harsh than a plain minor chord, and widely used in jazz, neo-soul, R&B, and modern worship. The Am7 and Em7 open shapes are essential beginner chords. The moveable Am7 barre shape (using the A-string root) covers all 12 keys efficiently.

Formula: Root + 3 + 4 + 3 semitones | Root, Min 3rd, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
Cm7	**x-3-5-3-4-3**	**C Eb G Bb**	*Barre 3rd — Am7 shape*
Dbm7	**x-4-6-4-5-4**	**Db E Ab B**	*Barre 4th — Am7 shape*
Dm7	**x-x-0-2-1-1**	**D F A C**	*Open Dm7 — mini barre strings 1-2*
Ebm7	**x-6-8-6-7-6**	**Eb Gb Bb Db**	*Barre 6th — Am7 shape*
Em7	**0-2-2-0-3-0**	**E G B D**	*Open Em7 — add pinky on B string*
Fm7	**1-3-3-1-4-1**	**F Ab C Eb**	*Barre 1st — Em7 shape*
Gbm7	**2-4-4-2-5-2**	**Gb A Db E**	*Barre 2nd — Em7 shape*
Gm7	**3-5-3-3-3-3**	**G Bb D F**	*Barre 3rd — full Gm7*
Abm7	**4-6-4-4-4-4**	**Ab B Eb Gb**	*Barre 4th — Em7 shape*
Am7	**x-0-2-0-1-0**	**A C E G**	*Open Am7 — one of most used chords*
Bbm7	**x-1-3-1-2-1**	**Bb Db F Ab**	*Barre 1st — Am7 shape*
Bm7	**x-2-4-2-3-2**	**B D F# A**	*Barre 2nd — Am7 shape*

27. Minor Major 7th (mMaj7) — Guitar Voicings

The Minor Major 7th chord is one of the most dramatic sounds in harmony — dark, cinematic, and mysterious. On guitar it is slightly tricky to finger but very rewarding. The AmMaj7 open shape is particularly beautiful and is a great starting point. This chord comes from the harmonic minor scale and appears in film scores and advanced jazz.

Formula: Root + 3 + 4 + 4 semitones | Root, Min 3rd, P 5th, Maj 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
CmMaj7	x-3-5-4-4-3	C Eb G B	*Barre 3rd — dramatic sound*
DbmMaj7	x-4-6-5-5-4	Db E Ab C	*Barre 4th shape*
DmMaj7	x-x-0-2-2-1	D F A C#	*Open — lift pinky vs Dm7*
EbmMaj7	x-6-8-7-7-6	Eb Gb Bb D	*Barre 6th shape*
EmMaj7	0-2-2-0-0-0	E G B D#	*Open Em + raise 7th one fret*
FmMaj7	1-3-3-1-1-0	F Ab C E	*Barre 1st — high e open*
GbmMaj7	2-4-4-2-2-1	Gb A Db F	*Barre 2nd shape*
GmMaj7	3-5-5-3-3-2	G Bb D F#	*Barre 3rd — cinematic feel*
AbmMaj7	4-6-6-4-4-3	Ab B Eb G	*Barre 4th shape*
AmMaj7	x-0-2-1-1-0	A C E G#	*Open — gorgeous harmonic minor sound*
BbmMaj7	x-1-3-2-2-1	Bb Db F A	*Barre 1st shape*
BmMaj7	x-2-4-3-3-2	B D F# A#	*Barre 2nd shape*

28. Half Diminished / m7b5 — Guitar Voicings

The Half Diminished (m7b5) is the standard ii chord in a minor ii-V-i progression in jazz (e.g. Bm7b5 - E7 - Am). On guitar the moveable shape is relatively compact once learned. The open Am7b5 shape is a good entry point. This chord also appears in film music whenever a sense of unease or instability is needed.

Formula: Root + 3 + 3 + 4 semitones | Root, Min 3rd, Dim 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
Cm7b5	x-3-4-3-4-3	C Eb Gb Bb	*Barre 3rd — tense jazz voicing*
Dbm7b5	x-4-5-4-5-4	Db E G B	*Barre 4th shape*
Dm7b5	x-x-0-1-1-1	D F Ab C	*Open — common in Am ii-V-i*
Ebm7b5	x-6-7-6-7-6	Eb Gb A Db	*Barre 6th shape*

Em7b5	**0-1-2-0-3-0**	**E G Bb D**	*Open position — unusual but works*
Fm7b5	**1-2-3-1-4-1**	**F Ab B Eb**	*Barre 1st shape*
Gbm7b5	**2-3-4-2-5-2**	**Gb A C E**	*Barre 2nd shape*
Gm7b5	**3-4-5-3-6-3**	**G Bb Db F**	*Barre 3rd shape*
Abm7b5	**x-4-5-4-4-4**	**Ab B D Gb**	*Barre 4th — compact voicing*
Am7b5	**x-0-1-0-1-0**	**A C Eb G**	*Open Am7b5 — easy jazz shape*
Bbm7b5	**x-1-2-1-2-1**	**Bb Db E Ab**	*Barre 1st shape*
Bm7b5	**x-2-3-2-3-2**	**B D F A**	*Barre 2nd — jazz minor ii chord*

29. Diminished 7th (dim7) — Guitar Voicings

The Diminished 7th chord is fully symmetrical — every note is exactly 3 semitones apart. This means any dim7 shape can be slid up or down 3 frets and it is still a valid dim7 chord (just a different inversion). On guitar, one moveable shape covers 4 chord names at once. It creates maximum tension and is used as a chromatic passing chord in gospel, jazz, and classical music.

Formula: Root + 3 + 3 + 3 semitones | Root, Min 3rd, Dim 5th, Dim 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
Cdim7	**x-3-4-2-4-2**	**C Eb Gb A**	*Symmetrical — same as Ebdim7*
Dbdim7	**x-4-5-3-5-3**	**Db E G Bb**	*Barre 4th — same as Edim7*
Ddim7	**x-5-6-4-6-4**	**D F Ab B**	*Barre 5th — same as Fdim7*
Ebdim7	**x-3-4-2-4-2**	**Eb Gb A C**	*Same shape as Cdim7*
Edim7	**0-1-2-0-2-0**	**E G Bb Db**	*Open Edim7 — dramatic sound*
Fdim7	**1-2-3-1-3-1**	**F Ab B D**	*Barre 1st — same as Abdim7*
Gbdim7	**2-3-4-2-4-2**	**Gb A C Eb**	*Barre 2nd — same as Adim7*
Gdim7	**3-4-5-3-5-3**	**G Bb Db E**	*Barre 3rd — same as Bbdim7*
Abdim7	**1-2-3-1-3-1**	**Ab B D F**	*Same shape as Fdim7*
Adim7	**x-0-1-2-1-2**	**A C Eb Gb**	*Open Adim7 — gothic feel*
Bbdim7	**x-1-2-3-2-3**	**Bb Db E G**	*Barre 1st — compact shape*
Bdim7	**x-2-3-4-3-4**	**B D F Ab**	*Barre 2nd — jazz passing chord*

30. Sus2 — Guitar Voicings

Sus2 chords on guitar have an open, floating sound that is instantly recognisable in modern worship music. Because the 3rd is replaced by the 2nd, these chords work over both major and minor harmonies. The Asus2, Gsus2, Dsus2, and Esus2 open shapes are among the most commonly used chords in contemporary worship guitar.

Formula: Root + 2 + 5 semitones | Root, Maj 2nd, P 5th (3rd omitted)

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
Csus2	x-3-3-0-1-3	C D G	*Dreamy open voicing*
Dbsus2	x-4-4-1-2-4	Db Eb Ab	*Barre 1st — sus2 shape*
Dsus2	x-x-0-2-3-0	D E A	*Open D — lift 3rd finger of Dsus4*
Ebsus2	x-6-6-3-4-6	Eb F Bb	*Barre 3rd — sus2 shape*
Esus2	0-2-4-4-0-0	E F# B	*Open E sus2 — airy sound*
Fsus2	x-8-8-5-6-8	F G C	*Barre 8th — or capo 5 use Csus2*
Gbsus2	2-4-4-1-2-2	Gb Ab Db	*Barre 2nd shape*
Gsus2	3-0-0-0-3-3	G A D	*Open G sus2 — popular worship voicing*
Absus2	x-6-8-8-4-4	Ab Bb Eb	*Barre 4th — compact sus2*
Asus2	x-0-2-2-0-0	A B E	*Open Asus2 — very common worship chord*
Bbsus2	x-1-3-3-1-1	Bb C F	*Barre 1st shape*
Bsus2	x-2-4-4-2-2	B C# F#	*Barre 2nd shape*

31. Sus4 — Guitar Voicings

Sus4 chords create a feeling of anticipation and suspension that wants to resolve to the major chord. On guitar they are easy to form from standard chord shapes — usually just add one finger or shift the 3rd up one fret. Gsus4, Dsus4, Asus4, and Esus4 are essential open shapes. The sus4 sound appears in rock intros, worship builds, and jazz cadences.

Formula: Root + 5 + 2 semitones | Root, P 4th, P 5th (3rd omitted)

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
Csus4	x-3-3-0-1-1	C F G	*Open C shape — barre strings 1-2*
Dbsus4	x-4-6-6-2-2	Db Gb Ab	*Barre 2nd fret partial*
Dsus4	x-x-0-2-3-3	D G A	*Open Dsus4 — add pinky on high e*
Ebsus4	x-6-8-8-4-4	Eb Ab Bb	*Barre 6th — Asus4 shape*

Esus4	**0-2-2-2-0-0**	**E A B**	*Open Esus4 — very easy open shape*
Fsus4	**1-3-3-3-1-1**	**F Bb C**	*Barre 1st — Esus4 shape*
Gbsus4	**2-4-4-4-2-2**	**Gb B Db**	*Barre 2nd — Esus4 shape*
Gsus4	**3-3-0-0-1-3**	**G C D**	*Open Gsus4 — beautiful sound*
Absus4	**4-6-6-6-4-4**	**Ab Db Eb**	*Barre 4th — Esus4 shape*
Asus4	**x-0-2-2-3-0**	**A D E**	*Open Asus4 — just add pinky*
Bbsus4	**x-1-3-3-4-1**	**Bb Eb F**	*Barre 1st — Asus4 shape*
Bsus4	**x-2-4-4-5-2**	**B E F#**	*Barre 2nd — Asus4 shape*

32. 7sus2 — Guitar Voicings

The 7sus2 chord adds a minor 7th to the sus2 for a lush, modern sound popular in contemporary worship (Hillsong, Elevation Worship) and neo-soul. On guitar these voicings work beautifully when the bass player holds the root while you play the upper structure — the G7sus2 and A7sus2 open shapes are especially popular.

Formula: Root + 2 + 5 + 3 semitones | Root, Maj 2nd, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
C7sus2	**x-3-3-3-1-3**	**C D G Bb**	*Open C7sus2 — lush worship sound*
Db7sus2	**x-4-4-4-2-4**	**Db Eb Ab B**	*Barre 4th shape*
D7sus2	**x-0-0-2-1-0**	**D E A C**	*Open — dreamy neo-soul voicing*
Eb7sus2	**x-6-6-6-4-6**	**Eb F Bb Db**	*Barre 6th shape*
E7sus2	**0-2-4-4-0-3**	**E F# B D**	*Open E7sus2*
F7sus2	**1-3-3-3-1-4**	**F G C Eb**	*Barre 1st shape*
Gb7sus2	**2-4-4-4-2-5**	**Gb Ab Db E**	*Barre 2nd shape*
G7sus2	**3-0-0-0-1-1**	**G A D F**	*Open G7sus2 — very popular in worship*
Ab7sus2	**4-6-6-6-4-7**	**Ab Bb Eb Gb**	*Barre 4th shape*
A7sus2	**x-0-2-0-0-0**	**A B E G**	*Open A7sus2 — float above the bass*
Bb7sus2	**x-1-3-1-1-1**	**Bb C F Ab**	*Barre 1st — smooth jazz feel*
B7sus2	**x-2-4-2-2-2**	**B C# F# A**	*Barre 2nd shape*

33. 7sus4 — Guitar Voicings

The 7sus4 is the classic suspended dominant chord — it sounds like a dominant 7th but with the 4th instead of the 3rd, giving it a floaty, unresolved feel. It appears constantly in jazz (as a modern V chord), gospel, and pop. The G7sus4 and D7sus4 open shapes are essential. Resolution path: 7sus4 → 7 → I (e.g. G7sus4 → G7 → C).

Formula: Root + 5 + 2 + 3 semitones | Root, P 4th, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 → Str5 → Str4 → Str3 → Str2 → Str1

Chord	Fret Shape (Str6→Str1)	Notes in Chord	Tip
C7sus4	x-3-3-3-1-1	C F G Bb	*Classic suspended dominant*
Db7sus4	x-4-4-4-2-2	Db Gb Ab B	*Barre 4th — Asus4+7 shape*
D7sus4	x-0-0-2-1-1	D G A C	*Open D7sus4 — mini barre*
Eb7sus4	x-6-6-6-4-4	Eb Ab Bb Db	*Barre 6th shape*
E7sus4	0-2-2-2-3-0	E A B D	*Open E7sus4 — jazz V chord*
F7sus4	1-3-3-3-4-1	F Bb C Eb	*Barre 1st — E7sus4 shape*
Gb7sus4	2-4-4-4-5-2	Gb B Db E	*Barre 2nd shape*
G7sus4	3-3-0-0-1-1	G C D F	*Open G7sus4 — resolves to G7→C*
Ab7sus4	4-6-6-6-7-4	Ab Db Eb Gb	*Barre 4th shape*
A7sus4	x-0-2-0-3-0	A D E G	*Open A7sus4 — add pinky on B string*
Bb7sus4	x-1-3-1-4-1	Bb Eb F Ab	*Barre 1st — A7sus4 shape*
B7sus4	x-2-4-2-5-2	B E F# A	*Barre 2nd — A7sus4 shape*

34. Major 7th (Maj7) Inversions — Keyboard

A 4-note chord has 4 positions: Root Position (root in bass), 1st Inversion (3rd in bass), 2nd Inversion (5th in bass), and 3rd Inversion (7th in bass). On the keyboard, inverting a Maj7 chord gives you lush, close voicings that are ideal for smooth voice leading in jazz and worship. The notes are always the same — only the lowest note changes. Notes read low to high, left to right.

Root Pos = R-3-5-7 | 1st Inv = 3-5-7-R | 2nd Inv = 5-7-R-3 | 3rd Inv = 7-R-3-5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
CMaj7	C — E — G — B	E — G — B — C	G — B — C — E	B — C — E — G
DbMaj7	Db — F — Ab — C	F — Ab — C — Db	Ab — C — Db — F	C — Db — F — Ab
DMaj7	D — Gb — A — Db	Gb — A — Db — D	A — Db — D — Gb	Db — D — Gb — A
EbMaj7	Eb — G — Bb — D	G — Bb — D — Eb	Bb — D — Eb — G	D — Eb — G — Bb

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
EMaj7	E — Ab — B — Eb	Ab — B — Eb — E	B — Eb — E — Ab	Eb — E — Ab — B
FMaj7	F — A — C — E	A — C — E — F	C — E — F — A	E — F — A — C
GbMaj7	Gb — Bb — Db — F	Bb — Db — F — Gb	Db — F — Gb — Bb	F — Gb — Bb — Db
GMaj7	G — B — D — Gb	B — D — Gb — G	D — Gb — G — B	Gb — G — B — D
AbMaj7	Ab — C — Eb — G	C — Eb — G — Ab	Eb — G — Ab — C	G — Ab — C — Eb
AMaj7	A — Db — E — Ab	Db — E — Ab — A	E — Ab — A — Db	Ab — A — Db — E
BbMaj7	Bb — D — F — A	D — F — A — Bb	F — A — Bb — D	A — Bb — D — F
BMaj7	B — Eb — Gb — Bb	Eb — Gb — Bb — B	Gb — Bb — B — Eb	Bb — B — Eb — Gb

35. Dominant 7th (7) Inversions — Keyboard

Inverting a Dominant 7th chord creates strong voice-leading motion. The 3rd inversion (7th in bass) is especially useful — it has the flattest 7th in the bass, which creates a very smooth descending bassline. Dominant 7th inversions are essential for jazz walking bass lines and gospel keyboard playing. Each inversion has a unique sound and function.

Root Pos = R-3-5-b7 | 1st Inv = 3-5-b7-R | 2nd Inv = 5-b7-R-3 | 3rd Inv = b7-R-3-5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
C7	C — E — G — Bb	E — G — Bb — C	G — Bb — C — E	Bb — C — E — G
Db7	Db — F — Ab — B	F — Ab — B — Db	Ab — B — Db — F	B — Db — F — Ab
D7	D — Gb — A — C	Gb — A — C — D	A — C — D — Gb	C — D — Gb — A
Eb7	Eb — G — Bb — Db	G — Bb — Db — Eb	Bb — Db — Eb — G	Db — Eb — G — Bb
E7	E — Ab — B — D	Ab — B — D — E	B — D — E — Ab	D — E — Ab — B
F7	F — A — C — Eb	A — C — Eb — F	C — Eb — F — A	Eb — F — A — C
Gb7	Gb — Bb — Db — E	Bb — Db — E — Gb	Db — E — Gb — Bb	E — Gb — Bb — Db
G7	G — B — D — F	B — D — F — G	D — F — G — B	F — G — B — D
Ab7	Ab — C — Eb — Gb	C — Eb — Gb — Ab	Eb — Gb — Ab — C	Gb — Ab — C — Eb
A7	A — Db — E — G	Db — E — G — A	E — G — A — Db	G — A — Db — E
Bb7	Bb — D — F — Ab	D — F — Ab — Bb	F — Ab — Bb — D	Ab — Bb — D — F
B7	B — Eb — Gb — A	Eb — Gb — A — B	Gb — A — B — Eb	A — B — Eb — Gb

36. Minor 7th (m7) Inversions — Keyboard

Minor 7th chord inversions give a smooth, soulful quality on the keyboard. The 1st inversion (minor 3rd in bass) is particularly warm-sounding and often used as a substitute for a plain minor chord. In neo-soul and contemporary worship, the 2nd and 3rd inversions of m7 chords create flowing inner-voice movement that adds depth to any progression.

Root Pos = R-b3-5-b7 | 1st Inv = b3-5-b7-R | 2nd Inv = 5-b7-R-b3 | 3rd Inv = b7-R-b3-5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
Cm7	C — Eb — G — Bb	Eb — G — Bb — C	G — Bb — C — Eb	Bb — C — Eb — G
Dbm7	Db — E — Ab — B	E — Ab — B — Db	Ab — B — Db — E	B — Db — E — Ab
Dm7	D — F — A — C	F — A — C — D	A — C — D — F	C — D — F — A
Ebm7	Eb — Gb — Bb — Db	Gb — Bb — Db — Eb	Bb — Db — Eb — Gb	Db — Eb — Gb — Bb
Em7	E — G — B — D	G — B — D — E	B — D — E — G	D — E — G — B
Fm7	F — Ab — C — Eb	Ab — C — Eb — F	C — Eb — F — Ab	Eb — F — Ab — C
Gbm7	Gb — A — Db — E	A — Db — E — Gb	Db — E — Gb — A	E — Gb — A — Db
Gm7	G — Bb — D — F	Bb — D — F — G	D — F — G — Bb	F — G — Bb — D
Abm7	Ab — B — Eb — Gb	B — Eb — Gb — Ab	Eb — Gb — Ab — B	Gb — Ab — B — Eb
Am7	A — C — E — G	C — E — G — A	E — G — A — C	G — A — C — E
Bbm7	Bb — Db — F — Ab	Db — F — Ab — Bb	F — Ab — Bb — Db	Ab — Bb — Db — F
Bm7	B — D — Gb — A	D — Gb — A — B	Gb — A — B — D	A — B — D — Gb

37. Minor Major 7th (mMaj7) Inversions — Keyboard

The Minor Major 7th chord is from the harmonic minor scale and has a distinctive dark, cinematic colour. Its inversions are rare in pop but widely used in film scoring and advanced jazz. The 2nd inversion (5th in bass) sounds particularly striking. These inversions are excellent for studying voice leading in minor keys.

Root Pos = R-b3-5-7 | 1st Inv = b3-5-7-R | 2nd Inv = 5-7-R-b3 | 3rd Inv = 7-R-b3-5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
CmMaj7	C — Eb — G — B	Eb — G — B — C	G — B — C — Eb	B — C — Eb — G
DbmMaj7	Db — E — Ab — C	E — Ab — C — Db	Ab — C — Db — E	C — Db — E — Ab
DmMaj7	D — F — A — Db	F — A — Db — D	A — Db — D — F	Db — D — F — A
EbmMaj7	Eb — Gb — Bb — D	Gb — Bb — D — Eb	Bb — D — Eb — Gb	D — Eb — Gb — Bb
EmMaj7	E — G — B — Eb	G — B — Eb — E	B — Eb — E — G	Eb — E — G — B

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
FmMaj7	F — Ab — C — E	Ab — C — E — F	C — E — F — Ab	E — F — Ab — C
GbmMaj7	Gb — A — Db — F	A — Db — F — Gb	Db — F — Gb — A	F — Gb — A — Db
GmMaj7	G — Bb — D — Gb	Bb — D — Gb — G	D — Gb — G — Bb	Gb — G — Bb — D
AbmMaj7	Ab — B — Eb — G	B — Eb — G — Ab	Eb — G — Ab — B	G — Ab — B — Eb
AmMaj7	A — C — E — Ab	C — E — Ab — A	E — Ab — A — C	Ab — A — C — E
BbmMaj7	Bb — Db — F — A	Db — F — A — Bb	F — A — Bb — Db	A — Bb — Db — F
BmMaj7	B — D — Gb — Bb	D — Gb — Bb — B	Gb — Bb — B — D	Bb — B — D — Gb

38. Half Diminished / m7b5 Inversions — Keyboard

The Half Diminished chord (m7b5) is the ii chord in minor key jazz progressions (ii-V-i). Its inversions on keyboard allow smooth voice leading into the dominant V chord. The 3rd inversion (minor 7th in bass) creates a particularly tense and interesting colour. These inversions are a staple of jazz harmony and sophisticated worship playing.

Root Pos = R-b3-b5-b7 | 1st Inv = b3-b5-b7-R | 2nd Inv = b5-b7-R-b3 | 3rd Inv = b7-R-b3-b5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
Cm7b5	C — Eb — Gb — Bb	Eb — Gb — Bb — C	Gb — Bb — C — Eb	Bb — C — Eb — Gb
Dbm7b5	Db — E — G — B	E — G — B — Db	G — B — Db — E	B — Db — E — G
Dm7b5	D — F — Ab — C	F — Ab — C — D	Ab — C — D — F	C — D — F — Ab
Ebm7b5	Eb — Gb — A — Db	Gb — A — Db — Eb	A — Db — Eb — Gb	Db — Eb — Gb — A
Em7b5	E — G — Bb — D	G — Bb — D — E	Bb — D — E — G	D — E — G — Bb
Fm7b5	F — Ab — B — Eb	Ab — B — Eb — F	B — Eb — F — Ab	Eb — F — Ab — B
Gbm7b5	Gb — A — C — E	A — C — E — Gb	C — E — Gb — A	E — Gb — A — C
Gm7b5	G — Bb — Db — F	Bb — Db — F — G	Db — F — G — Bb	F — G — Bb — Db
Abm7b5	Ab — B — D — Gb	B — D — Gb — Ab	D — Gb — Ab — B	Gb — Ab — B — D
Am7b5	A — C — Eb — G	C — Eb — G — A	Eb — G — A — C	G — A — C — Eb
Bbm7b5	Bb — Db — E — Ab	Db — E — Ab — Bb	E — Ab — Bb — Db	Ab — Bb — Db — E
Bm7b5	B — D — F — A	D — F — A — B	F — A — B — D	A — B — D — F

39. Diminished 7th (dim7) Inversions — Keyboard

The Diminished 7th chord is fully symmetrical — all intervals are equal minor 3rds (3 semitones each). This means every inversion of a dim7 chord is itself another dim7 chord name. For example, Cdim7 in 1st inversion = Ebdim7 in root position (same notes). This symmetry makes dim7 inversions very efficient on keyboard — one shape covers 4 chords. The dim7 is widely used as a chromatic passing chord.

Root Pos = R-b3-b5-bb7 | 1st Inv = b3-b5-bb7-R | 2nd Inv = b5-bb7-R-b3 | 3rd Inv = bb7-R-b3-b5 (all same note-set, enharmonically)

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
Cdim7	C — Eb — Gb — A	Eb — Gb — A — C	Gb — A — C — Eb	A — C — Eb — Gb
Dbdim7	Db — E — G — Bb	E — G — Bb — Db	G — Bb — Db — E	Bb — Db — E — G
Ddim7	D — F — Ab — B	F — Ab — B — D	Ab — B — D — F	B — D — F — Ab
Ebdim7	Eb — Gb — A — C	Gb — A — C — Eb	A — C — Eb — Gb	C — Eb — Gb — A
Edim7	E — G — Bb — Db	G — Bb — Db — E	Bb — Db — E — G	Db — E — G — Bb
Fdim7	F — Ab — B — D	Ab — B — D — F	B — D — F — Ab	D — F — Ab — B
Gbdim7	Gb — A — C — Eb	A — C — Eb — Gb	C — Eb — Gb — A	Eb — Gb — A — C
Gdim7	G — Bb — Db — E	Bb — Db — E — G	Db — E — G — Bb	E — G — Bb — Db
Abdim7	Ab — B — D — F	B — D — F — Ab	D — F — Ab — B	F — Ab — B — D
Adim7	A — C — Eb — Gb	C — Eb — Gb — A	Eb — Gb — A — C	Gb — A — C — Eb
Bbdim7	Bb — Db — E — G	Db — E — G — Bb	E — G — Bb — Db	G — Bb — Db — E
Bdim7	B — D — F — Ab	D — F — Ab — B	F — Ab — B — D	Ab — B — D — F

40. Sus2 Inversions — Keyboard

Sus2 is a 3-note chord (Root, Maj 2nd, Perfect 5th), so it has 3 positions. The 1st inversion places the 2nd (Maj 2nd) in the bass, and the 2nd inversion places the 5th in the bass. Sus2 inversions on keyboard produce open, floating sounds with no strong tonal direction — they work beautifully as sustained pads and for ambiguous harmonic colour in worship music.

Root Pos = R-2-5 | 1st Inv (2nd in bass) = 2-5-R | 2nd Inv (5th in bass) = 5-R-2

Chord	Root Position — (Root in bass)	1st Inversion — (2nd or 4th in bass)	2nd Inversion — (5th in bass)	Slash Notes
Csus2	C — D — G	D — G — C	G — C — D	D / G
Dbsus2	Db — Eb — Ab	Eb — Ab — Db	Ab — Db — Eb	Eb / Ab
Dsus2	D — E — A	E — A — D	A — D — E	E / A
Ebsus2	Eb — F — Bb	F — Bb — Eb	Bb — Eb — F	F / Bb
Esus2	E — Gb — B	Gb — B — E	B — E — Gb	Gb / B

Chord	Root Position — (Root in bass)	1st Inversion — (2nd or 4th in bass)	2nd Inversion — (5th in bass)	Slash Notes
Fsus2	F — G — C	G — C — F	C — F — G	G / C
Gbsus2	Gb — Ab — Db	Ab — Db — Gb	Db — Gb — Ab	Ab / Db
Gsus2	G — A — D	A — D — G	D — G — A	A / D
Absus2	Ab — Bb — Eb	Bb — Eb — Ab	Eb — Ab — Bb	Bb / Eb
Asus2	A — B — E	B — E — A	E — A — B	B / E
Bbsus2	Bb — C — F	C — F — Bb	F — Bb — C	C / F
Bsus2	B — Db — Gb	Db — Gb — B	Gb — B — Db	Db / Gb

41. Sus4 Inversions — Keyboard

Sus4 is a 3-note chord (Root, Perfect 4th, Perfect 5th). Its 1st inversion places the 4th in the bass and 2nd inversion places the 5th. The 4th in the bass gives a strong feeling of forward motion — it almost sounds like a plagal (IV→I) movement. Sus4 inversions are excellent for creating gentle movement in a sustained keyboard part without committing to a full chord change.

Root Pos = R-4-5 | 1st Inv (4th in bass) = 4-5-R | 2nd Inv (5th in bass) = 5-R-4

Chord	Root Position — (Root in bass)	1st Inversion — (2nd or 4th in bass)	2nd Inversion — (5th in bass)	Slash Notes
Csus4	C — F — G	F — G — C	G — C — F	F / G
Dbsus4	Db — Gb — Ab	Gb — Ab — Db	Ab — Db — Gb	Gb / Ab
Dsus4	D — G — A	G — A — D	A — D — G	G / A
Ebsus4	Eb — Ab — Bb	Ab — Bb — Eb	Bb — Eb — Ab	Ab / Bb
Esus4	E — A — B	A — B — E	B — E — A	A / B
Fsus4	F — Bb — C	Bb — C — F	C — F — Bb	Bb / C
Gbsus4	Gb — B — Db	B — Db — Gb	Db — Gb — B	B / Db
Gsus4	G — C — D	C — D — G	D — G — C	C / D
Absus4	Ab — Db — Eb	Db — Eb — Ab	Eb — Ab — Db	Db / Eb
Asus4	A — D — E	D — E — A	E — A — D	D / E
Bbsus4	Bb — Eb — F	Eb — F — Bb	F — Bb — Eb	Eb / F
Bsus4	B — E — Gb	E — Gb — B	Gb — B — E	E / Gb

42. 7sus2 Inversions — Keyboard

The 7sus2 chord has 4 notes (Root, Maj 2nd, Perfect 5th, Minor 7th), so it has 4 inversion positions. On keyboard, 7sus2 inversions are especially popular in contemporary worship as they create lush, open-voiced pads. The 3rd inversion (minor 7th in bass) gives a very modern, floaty sound. Try voicing these with left hand on the bass note and right hand on the upper 3 notes.

Root Pos = R-2-5-b7 | 1st Inv = 2-5-b7-R | 2nd Inv = 5-b7-R-2 | 3rd Inv = b7-R-2-5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
C7sus2	C — D — G — Bb	D — G — Bb — C	G — Bb — C — D	Bb — C — D — G
Db7sus2	Db — Eb — Ab — B	Eb — Ab — B — Db	Ab — B — Db — Eb	B — Db — Eb — Ab
D7sus2	D — E — A — C	E — A — C — D	A — C — D — E	C — D — E — A
Eb7sus2	Eb — F — Bb — Db	F — Bb — Db — Eb	Bb — Db — Eb — F	Db — Eb — F — Bb
E7sus2	E — Gb — B — D	Gb — B — D — E	B — D — E — Gb	D — E — Gb — B
F7sus2	F — G — C — Eb	G — C — Eb — F	C — Eb — F — G	Eb — F — G — C
Gb7sus2	Gb — Ab — Db — E	Ab — Db — E — Gb	Db — E — Gb — Ab	E — Gb — Ab — Db
G7sus2	G — A — D — F	A — D — F — G	D — F — G — A	F — G — A — D
Ab7sus2	Ab — Bb — Eb — Gb	Bb — Eb — Gb — Ab	Eb — Gb — Ab — Bb	Gb — Ab — Bb — Eb
A7sus2	A — B — E — G	B — E — G — A	E — G — A — B	G — A — B — E
Bb7sus2	Bb — C — F — Ab	C — F — Ab — Bb	F — Ab — Bb — C	Ab — Bb — C — F
B7sus2	B — Db — Gb — A	Db — Gb — A — B	Gb — A — B — Db	A — B — Db — Gb

43. 7sus4 Inversions — Keyboard

The 7sus4 chord has 4 notes (Root, Perfect 4th, Perfect 5th, Minor 7th). It is the classic suspended dominant — it sounds like it wants to resolve. On keyboard, the 3rd inversion (minor 7th in bass) is very common in jazz and creates a strong pull toward resolution. The 1st inversion (4th in bass) sounds like a IV chord with added 7th — extremely useful for gospel keyboard style.

Root Pos = R-4-5-b7 | 1st Inv = 4-5-b7-R | 2nd Inv = 5-b7-R-4 | 3rd Inv = b7-R-4-5

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
C7sus4	C — F — G — Bb	F — G — Bb — C	G — Bb — C — F	Bb — C — F — G
Db7sus4	Db — Gb — Ab — B	Gb — Ab — B — Db	Ab — B — Db — Gb	B — Db — Gb — Ab
D7sus4	D — G — A — C	G — A — C — D	A — C — D — G	C — D — G — A
Eb7sus4	Eb — Ab — Bb — Db	Ab — Bb — Db — Eb	Bb — Db — Eb — Ab	Db — Eb — Ab — Bb

Chord	Root Position — (Root in bass)	1st Inversion — (3rd in bass)	2nd Inversion — (5th in bass)	3rd Inversion — (7th in bass)
E7sus4	E — A — B — D	A — B — D — E	B — D — E — A	D — E — A — B
F7sus4	F — Bb — C — Eb	Bb — C — Eb — F	C — Eb — F — Bb	Eb — F — Bb — C
Gb7sus4	Gb — B — Db — E	B — Db — E — Gb	Db — E — Gb — B	E — Gb — B — Db
G7sus4	G — C — D — F	C — D — F — G	D — F — G — C	F — G — C — D
Ab7sus4	Ab — Db — Eb — Gb	Db — Eb — Gb — Ab	Eb — Gb — Ab — Db	Gb — Ab — Db — Eb
A7sus4	A — D — E — G	D — E — G — A	E — G — A — D	G — A — D — E
Bb7sus4	Bb — Eb — F — Ab	Eb — F — Ab — Bb	F — Ab — Bb — Eb	Ab — Bb — Eb — F
B7sus4	B — E — Gb — A	E — Gb — A — B	Gb — A — B — E	A — B — E — Gb

44. Major 7th (Maj7) Inversions — Guitar

Maj7 chord inversions on guitar allow for smooth, voice-led movement without changing hand position too drastically. The inversion is achieved by placing the target note on the lowest sounding string. 1st inversion (3rd in bass) creates a bright, lifted sound. 2nd inversion (5th in bass) sounds stable and open. 3rd inversion (Maj7th in bass) is the most jazzy and needs careful voicing. Slash notation (e.g. CMaj7/E) tells you which note is in the bass.

Root + 4 + 3 + 4 semitones | Root, Maj 3rd, P 5th, Maj 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
CMaj7	x-3-2-0-0-0	CMaj7/E → x-3-2-0-0-0	CMaj7/G → x-3-2-0-0-3	CMaj7/B → x-3-2-0-0-7
DbMaj7	x-4-3-1-1-1	DbMaj7/F → x-4-3-1-1-1	DbMaj7/Ab → x-4-3-1-1-4	DbMaj7/C → x-4-3-1-1-x
DMaj7	x-x-0-2-2-2	DMaj7/Gb → x-x-0-2-2-2	DMaj7/A → x-x-0-2-2-5	DMaj7/Db → x-x-0-2-2-x
EbMaj7	x-6-5-3-3-3	EbMaj7/G → x-6-5-3-3-3	EbMaj7/Bb → x-6-5-3-3-6	EbMaj7/D → x-6-5-3-3-x
EMaj7	0-2-1-1-0-0	EMaj7/Ab → 0-2-1-1-0-4	EMaj7/B → 0-2-1-1-0-7	EMaj7/Eb → 0-2-1-1-4-x
FMaj7	1-3-3-2-1-0	FMaj7/A → 1-3-3-2-1-5	FMaj7/C → 1-3-3-2-1-x	FMaj7/E → 1-3-3-2-1-0
GbMaj7	2-4-4-3-2-1	GbMaj7/Bb → 2-4-4-3-2-6	GbMaj7/Db → 2-4-4-3-2-x	GbMaj7/F → 2-4-4-3-2-1
GMaj7	3-2-0-0-0-2	GMaj7/B → 3-2-0-0-0-7	GMaj7/D → 3-2-0-0-3-x	GMaj7/Gb → 3-2-0-0-0-2

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
AbMaj7	4-6-5-5-4-4	AbMaj7/C → 4-6-5-5-1-x	AbMaj7/Eb → 4-6-5-5-4-x	AbMaj7/G → 4-6-5-5-4-3
AMaj7	x-0-2-1-2-0	AMaj7/Db → x-0-2-1-2-x	AMaj7/E → x-0-2-1-2-0	AMaj7/Ab → x-0-2-1-2-4
BbMaj7	x-1-3-2-3-1	BbMaj7/D → x-1-3-2-3-x	BbMaj7/F → x-1-3-2-3-1	BbMaj7/A → x-1-3-2-3-5
BMaj7	x-2-4-3-4-2	BMaj7/Eb → x-2-4-3-4-x	BMaj7/Gb → x-2-4-3-4-2	BMaj7/Bb → x-2-4-3-4-6

45. Dominant 7th (7) Inversions — Guitar

Dominant 7th inversions on guitar are widely used in blues and jazz for strong voice leading. The 1st inversion (major 3rd in bass) is called a "first inversion dominant" and creates a strong blues feel. The 3rd inversion (minor 7th in bass) is used as a passing chord. Knowing these inversions unlocks the ability to move through chord changes smoothly on the neck without big position shifts.

Root + 4 + 3 + 3 semitones | Root, Maj 3rd, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
C7	x-3-2-3-1-0	C7/E → x-3-2-3-1-0	C7/G → x-3-2-3-1-3	C7/Bb → x-3-2-3-1-6
Db7	x-4-3-4-2-1	Db7/F → x-4-3-4-2-1	Db7/Ab → x-4-3-4-2-4	Db7/B → x-4-3-4-2-7
D7	x-x-0-2-1-2	D7/Gb → x-x-0-2-1-2	D7/A → x-x-0-2-1-5	D7/C → x-x-0-2-1-x
Eb7	x-6-5-6-4-3	Eb7/G → x-6-5-6-4-3	Eb7/Bb → x-6-5-6-4-6	Eb7/Db → x-6-5-6-2-x
E7	0-2-0-1-0-0	E7/Ab → 0-2-0-1-0-4	E7/B → 0-2-0-1-0-7	E7/D → 0-2-0-1-3-x
F7	1-3-1-2-1-1	F7/A → 1-3-1-2-1-5	F7/C → 1-3-1-2-1-x	F7/Eb → 1-3-1-2-4-x
Gb7	2-4-2-3-2-2	Gb7/Bb → 2-4-2-3-2-6	Gb7/Db → 2-4-2-3-2-x	Gb7/E → 2-4-2-3-2-0
G7	3-2-0-0-0-1	G7/B → 3-2-0-0-0-7	G7/D → 3-2-0-0-3-x	G7/F → 3-2-0-0-0-1
Ab7	4-6-4-5-4-4	Ab7/C → 4-6-4-5-1-x	Ab7/Eb → 4-6-4-5-4-x	Ab7/Gb → 4-6-4-5-4-2

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
A7	x-0-2-0-2-0	A7/Db → x-0-2-0-2-x	A7/E → x-0-2-0-2-0	A7/G → x-0-2-0-2-3
Bb7	x-1-3-1-3-1	Bb7/D → x-1-3-1-3-x	Bb7/F → x-1-3-1-3-1	Bb7/Ab → x-1-3-1-3-4
B7	x-2-1-2-0-2	B7/Eb → x-2-1-2-4-x	B7/Gb → x-2-1-2-0-2	B7/A → x-2-1-2-0-5

46. Minor 7th (m7) Inversions — Guitar

Minor 7th inversions on guitar give a smooth, soulful sound and are essential for jazz and R&B guitar. The 1st inversion (minor 3rd in bass) sounds particularly warm and mellow. Minor 7th inversions work beautifully as ii chords in major-key progressions. Try connecting the inversions of the same chord in sequence for a smooth melodic bass line.

Root + 3 + 4 + 3 semitones | Root, Min 3rd, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
Cm7	x-3-5-3-4-3	Cm7/Eb → x-3-5-3-4-x	Cm7/G → x-3-5-3-4-3	Cm7/Bb → x-3-5-3-4-6
Dbm7	x-4-6-4-5-4	Dbm7/E → x-4-6-4-5-0	Dbm7/Ab → x-4-6-4-5-4	Dbm7/B → x-4-6-4-5-7
Dm7	x-x-0-2-1-1	Dm7/F → x-x-0-2-1-1	Dm7/A → x-x-0-2-1-5	Dm7/C → x-x-0-2-1-x
Ebm7	x-6-8-6-7-6	Ebm7/Gb → x-6-8-6-7-2	Ebm7/Bb → x-6-8-6-7-6	Ebm7/Db → x-6-8-6-2-x
Em7	0-2-2-0-3-0	Em7/G → 0-2-2-0-3-3	Em7/B → 0-2-2-0-3-7	Em7/D → 0-2-2-0-3-x
Fm7	1-3-3-1-4-1	Fm7/Ab → 1-3-3-1-4-4	Fm7/C → 1-3-3-1-1-x	Fm7/Eb → 1-3-3-1-4-x
Gbm7	2-4-4-2-5-2	Gbm7/A → 2-4-4-2-5-5	Gbm7/Db → 2-4-4-2-2-x	Gbm7/E → 2-4-4-2-5-0
Gm7	3-5-3-3-3-3	Gm7/Bb → 3-5-3-3-3-6	Gm7/D → 3-5-3-3-3-x	Gm7/F → 3-5-3-3-3-1
Abm7	4-6-4-4-4-4	Abm7/B → 4-6-4-4-4-7	Abm7/Eb → 4-6-4-4-4-x	Abm7/Gb → 4-6-4-4-4-2
Am7	x-0-2-0-1-0	Am7/C → x-0-2-0-1-x	Am7/E → x-0-2-0-1-0	Am7/G → x-0-2-0-1-3

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
Bbm7	x-1-3-1-2-1	Bbm7/Db → x-1-3-1-2-x	Bbm7/F → x-1-3-1-2-1	Bbm7/Ab → x-1-3-1-2-4
Bm7	x-2-4-2-3-2	Bm7/D → x-2-4-2-3-x	Bm7/Gb → x-2-4-2-3-2	Bm7/A → x-2-4-2-3-5

47. Minor Major 7th (mMaj7) Inversions — Guitar

Minor Major 7th inversions on guitar are from the harmonic minor scale and used in film music and advanced jazz. The 3rd inversion (major 7th in bass) is particularly dramatic and cinematic. These shapes require careful fingering — take them slowly. The minor Major 7th chord moving through its inversions creates a descending chromatic bass line (a famous device in classical and film music).

Root + 3 + 4 + 4 semitones | Root, Min 3rd, P 5th, Maj 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
CmMaj7	x-3-5-4-4-3	CmMaj7/Eb → x-3-5-4-4-x	CmMaj7/G → x-3-5-4-4-3	CmMaj7/B → x-3-5-4-4-7
DbmMaj7	x-4-6-5-5-4	DbmMaj7/E → x-4-6-5-5-0	DbmMaj7/Ab → x-4-6-5-5-4	DbmMaj7/C → x-4-6-5-1-x
DmMaj7	x-x-0-2-2-1	DmMaj7/F → x-x-0-2-2-1	DmMaj7/A → x-x-0-2-2-5	DmMaj7/Db → x-x-0-2-2-x
EbmMaj7	x-6-8-7-7-6	EbmMaj7/Gb → x-6-8-7-7-2	EbmMaj7/Bb → x-6-8-7-7-6	EbmMaj7/D → x-6-8-7-3-x
EmMaj7	0-2-2-0-0-0	EmMaj7/G → 0-2-2-0-0-3	EmMaj7/B → 0-2-2-0-0-7	EmMaj7/Eb → 0-2-2-0-4-x
FmMaj7	1-3-3-1-1-0	FmMaj7/Ab → 1-3-3-1-1-4	FmMaj7/C → 1-3-3-1-1-x	FmMaj7/E → 1-3-3-1-1-0
GbmMaj7	2-4-4-2-2-1	GbmMaj7/A → 2-4-4-2-2-5	GbmMaj7/Db → 2-4-4-2-2-x	GbmMaj7/F → 2-4-4-2-2-1
GmMaj7	3-5-5-3-3-2	GmMaj7/Bb → 3-5-5-3-3-6	GmMaj7/D → 3-5-5-3-3-x	GmMaj7/Gb → 3-5-5-3-3-2
AbmMaj7	4-6-6-4-4-3	AbmMaj7/B → 4-6-6-4-4-7	AbmMaj7/Eb → 4-6-6-4-4-x	AbmMaj7/G → 4-6-6-4-4-3
AmMaj7	x-0-2-1-1-0	AmMaj7/C → x-0-2-1-1-x	AmMaj7/E → x-0-2-1-1-0	AmMaj7/Ab → x-0-2-1-1-4
BbmMaj7	x-1-3-2-2-1	BbmMaj7/Db → x-1-3-2-2-x	BbmMaj7/F → x-1-3-2-2-1	BbmMaj7/A → x-1-3-2-2-5

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
BmMaj7	x-2-4-3-3-2	BmMaj7/D → x-2-4-3-3-x	BmMaj7/Gb → x-2-4-3-3-2	BmMaj7/Bb → x-2-4-3-3-6

48. Half Diminished / m7b5 Inversions — Guitar

Half Diminished inversions appear in jazz as the ii chord in minor ii-V-i progressions. On guitar, these inversions allow you to move through the ii-V-i pattern in one position. The 1st inversion (minor 3rd in bass) is less dissonant. The 3rd inversion (minor 7th in bass) is very tense and typically resolves down. Learning these inversions is essential for jazz guitar.

Root + 3 + 3 + 4 semitones | Root, Min 3rd, Dim 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
Cm7b5	x-3-4-3-4-3	Cm7b5/Eb → x-3-4-3-4-x	Cm7b5/Gb → x-3-4-3-4-2	Cm7b5/Bb → x-3-4-3-4-6
Dbm7b5	x-4-5-4-5-4	Dbm7b5/E → x-4-5-4-5-0	Dbm7b5/G → x-4-5-4-5-3	Dbm7b5/B → x-4-5-4-5-7
Dm7b5	x-x-0-1-1-1	Dm7b5/F → x-x-0-1-1-1	Dm7b5/Ab → x-x-0-1-1-4	Dm7b5/C → x-x-0-1-1-x
Ebm7b5	x-6-7-6-7-6	Ebm7b5/Gb → x-6-7-6-7-2	Ebm7b5/A → x-6-7-6-7-5	Ebm7b5/Db → x-6-7-6-2-x
Em7b5	0-1-2-0-3-0	Em7b5/G → 0-1-2-0-3-3	Em7b5/Bb → 0-1-2-0-3-6	Em7b5/D → 0-1-2-0-3-x
Fm7b5	1-2-3-1-4-1	Fm7b5/Ab → 1-2-3-1-4-4	Fm7b5/B → 1-2-3-1-4-7	Fm7b5/Eb → 1-2-3-1-4-x
Gbm7b5	2-3-4-2-5-2	Gbm7b5/A → 2-3-4-2-5-5	Gbm7b5/C → 2-3-4-2-1-x	Gbm7b5/E → 2-3-4-2-5-0
Gm7b5	3-4-5-3-6-3	Gm7b5/Bb → 3-4-5-3-6-6	Gm7b5/Db → 3-4-5-3-2-x	Gm7b5/F → 3-4-5-3-6-1
Abm7b5	x-4-5-4-4-4	Abm7b5/B → x-4-5-4-4-7	Abm7b5/D → x-4-5-4-3-x	Abm7b5/Gb → x-4-5-4-4-2
Am7b5	x-0-1-0-1-0	Am7b5/C → x-0-1-0-1-x	Am7b5/Eb → x-0-1-0-4-x	Am7b5/G → x-0-1-0-1-3
Bbm7b5	x-1-2-1-2-1	Bbm7b5/Db → x-1-2-1-2-x	Bbm7b5/E → x-1-2-1-2-0	Bbm7b5/Ab → x-1-2-1-2-4
Bm7b5	x-2-3-2-3-2	Bm7b5/D → x-2-3-2-3-x	Bm7b5/F → x-2-3-2-3-1	Bm7b5/A → x-2-3-2-3-5

49. Diminished 7th (dim7) Inversions — Guitar

Diminished 7th inversions on guitar are unique: because all intervals are equal (3 semitones), moving a dim7 shape up 3 frets gives the next inversion. One shape covers 4 chord names. For example, sliding Cdim7 (x-3-4-2-4-2) up 3 frets gives Ebdim7 in root position, which is the 1st inversion of Cdim7. This makes dim7 the most versatile chord shape on the guitar neck.

Root + 3 + 3 + 3 semitones | Root, Min 3rd, Dim 5th, Dim 7th (bb7)

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
Cdim7	x-3-4-2-4-2	Cdim7/Eb → x-3-4-2-4-x	Cdim7/Gb → x-3-4-2-4-2	Cdim7/A → x-3-4-2-4-5
Dbdim7	x-4-5-3-5-3	Dbdim7/E → x-4-5-3-5-0	Dbdim7/G → x-4-5-3-5-3	Dbdim7/Bb → x-4-5-3-5-6
Ddim7	x-5-6-4-6-4	Ddim7/F → x-5-6-4-6-1	Ddim7/Ab → x-5-6-4-6-4	Ddim7/B → x-5-6-4-6-7
Ebdim7	x-3-4-2-4-2	Ebdim7/Gb → x-3-4-2-4-2	Ebdim7/A → x-3-4-2-4-5	Ebdim7/C → x-3-4-2-1-x
Edim7	0-1-2-0-2-0	Edim7/G → 0-1-2-0-2-3	Edim7/Bb → 0-1-2-0-2-6	Edim7/Db → 0-1-2-0-2-x
Fdim7	1-2-3-1-3-1	Fdim7/Ab → 1-2-3-1-3-4	Fdim7/B → 1-2-3-1-3-7	Fdim7/D → 1-2-3-1-3-x
Gbdim7	2-3-4-2-4-2	Gbdim7/A → 2-3-4-2-4-5	Gbdim7/C → 2-3-4-2-1-x	Gbdim7/Eb → 2-3-4-2-4-x
Gdim7	3-4-5-3-5-3	Gdim7/Bb → 3-4-5-3-5-6	Gdim7/Db → 3-4-5-3-2-x	Gdim7/E → 3-4-5-3-5-0
Abdim7	1-2-3-1-3-1	Abdim7/B → 1-2-3-1-3-7	Abdim7/D → 1-2-3-1-3-x	Abdim7/F → 1-2-3-1-3-1
Adim7	x-0-1-2-1-2	Adim7/C → x-0-1-2-1-x	Adim7/Eb → x-0-1-2-4-x	Adim7/Gb → x-0-1-2-1-2
Bbdim7	x-1-2-3-2-3	Bbdim7/Db → x-1-2-3-2-x	Bbdim7/E → x-1-2-3-2-0	Bbdim7/G → x-1-2-3-2-3
Bdim7	x-2-3-4-3-4	Bdim7/D → x-2-3-4-3-x	Bdim7/F → x-2-3-4-3-1	Bdim7/Ab → x-2-3-4-3-4

50. Sus2 Inversions — Guitar

Sus2 chord inversions on guitar retain their open, floating quality in all positions. The 1st inversion (Maj 2nd in bass) creates a dreamy, unresolved sound. The 2nd inversion (5th in bass) is very open and airy. Sus2 inversions are beloved in contemporary worship because they work as pads — the bass player can hold the root while the guitarist plays an inversion for added colour.

Root + 2 + 5 semitones | Root, Maj 2nd, P 5th (3rd omitted)

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (2nd/4th = bass)	2nd Inversion — (5th = bass)	Slash Notes
Csus2	x-3-3-0-1-3	Csus2/D → x-3-3-0-3-x	Csus2/G → x-3-3-0-1-3	D / G
Dbsus2	x-4-4-1-2-4	Dbsus2/Eb → x-4-4-1-4-x	Dbsus2/Ab → x-4-4-1-2-4	Eb / Ab
Dsus2	x-x-0-2-3-0	Dsus2/E → x-x-0-2-3-0	Dsus2/A → x-x-0-2-3-5	E / A
Ebsus2	x-6-6-3-4-6	Ebsus2/F → x-6-6-3-4-1	Ebsus2/Bb → x-6-6-3-4-6	F / Bb
Esus2	0-2-4-4-0-0	Esus2/Gb → 0-2-4-4-0-2	Esus2/B → 0-2-4-4-0-7	Gb / B
Fsus2	x-8-8-5-6-8	Fsus2/G → x-8-8-5-6-3	Fsus2/C → x-8-8-5-1-x	G / C
Gbsus2	2-4-4-1-2-2	Gbsus2/Ab → 2-4-4-1-2-4	Gbsus2/Db → 2-4-4-1-2-x	Ab / Db
Gsus2	3-0-0-0-3-3	Gsus2/A → 3-0-0-0-3-5	Gsus2/D → 3-0-0-0-3-x	A / D
Absus2	x-6-8-8-4-4	Absus2/Bb → x-6-8-8-4-6	Absus2/Eb → x-6-8-8-4-x	Bb / Eb
Asus2	x-0-2-2-0-0	Asus2/B → x-0-2-2-0-7	Asus2/E → x-0-2-2-0-0	B / E
Bbsus2	x-1-3-3-1-1	Bbsus2/C → x-1-3-3-1-x	Bbsus2/F → x-1-3-3-1-1	C / F
Bsus2	x-2-4-4-2-2	Bsus2/Db → x-2-4-4-2-x	Bsus2/Gb → x-2-4-4-2-2	Db / Gb

51. Sus4 Inversions — Guitar

Sus4 inversions on guitar create strong harmonic anticipation. The 1st inversion (4th in the bass) sounds like a IV chord at the bottom — great for creating tension before resolving to the major shape. The 2nd inversion (5th in bass) is open and stable. Sus4 inversions are very common in rock and pop guitar, often used in arpeggiated patterns over sus4 → major resolutions.

Root + 5 + 2 semitones | Root, P 4th, P 5th (3rd omitted)

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (2nd/4th = bass)	2nd Inversion — (5th = bass)	Slash Notes
Csus4	x-3-3-0-1-1	Csus4/F → x-3-3-0-1-1	Csus4/G → x-3-3-0-1-3	F / G
Dbsus4	x-4-6-6-2-2	Dbsus4/Gb → x-4-6-6-2-2	Dbsus4/Ab → x-4-6-6-2-4	Gb / Ab
Dsus4	x-x-0-2-3-3	Dsus4/G → x-x-0-2-3-3	Dsus4/A → x-x-0-2-3-5	G / A
Ebsus4	x-6-8-8-4-4	Ebsus4/Ab → x-6-8-8-4-4	Ebsus4/Bb → x-6-8-8-4-6	Ab / Bb
Esus4	0-2-2-2-0-0	Esus4/A → 0-2-2-2-0-5	Esus4/B → 0-2-2-2-0-7	A / B
Fsus4	1-3-3-3-1-1	Fsus4/Bb → 1-3-3-3-1-6	Fsus4/C → 1-3-3-3-1-x	Bb / C
Gbsus4	2-4-4-4-2-2	Gbsus4/B → 2-4-4-4-2-7	Gbsus4/Db → 2-4-4-4-2-x	B / Db
Gsus4	3-3-0-0-1-3	Gsus4/C → 3-3-0-0-1-x	Gsus4/D → 3-3-0-0-3-x	C / D
Absus4	4-6-6-6-4-4	Absus4/Db → 4-6-6-6-2-x	Absus4/Eb → 4-6-6-6-4-x	Db / Eb
Asus4	x-0-2-2-3-0	Asus4/D → x-0-2-2-3-x	Asus4/E → x-0-2-2-3-0	D / E
Bbsus4	x-1-3-3-4-1	Bbsus4/Eb → x-1-3-3-4-x	Bbsus4/F → x-1-3-3-4-1	Eb / F
Bsus4	x-2-4-4-5-2	Bsus4/E → x-2-4-4-5-0	Bsus4/Gb → x-2-4-4-5-2	E / Gb

52. 7sus2 Inversions — Guitar

The 7sus2 chord (Root, Maj 2nd, P 5th, Min 7th) has 4 inversions on guitar. These lush, ambiguous voicings are a hallmark of contemporary worship guitar — think Hillsong and Elevation Worship. The 3rd inversion (minor 7th in bass) sounds particularly floaty and modern. These inversions work brilliantly when the bass player holds the root, freeing the guitarist to use any inversion for upper-structure colour.

Root + 2 + 5 + 3 semitones | Root, Maj 2nd, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
C7sus2	x-3-3-3-1-3	C7sus2/D → x-3-3-3-3-x	C7sus2/G → x-3-3-3-1-3	C7sus2/Bb → x-3-3-3-1-6
Db7sus2	x-4-4-4-2-4	Db7sus2/Eb → x-4-4-4-4-x	Db7sus2/Ab → x-4-4-4-2-4	Db7sus2/B → x-4-4-4-2-7
D7sus2	x-0-0-2-1-0	D7sus2/E → x-0-0-2-1-0	D7sus2/A → x-0-0-2-1-5	D7sus2/C → x-0-0-2-1-x
Eb7sus2	x-6-6-6-4-6	Eb7sus2/F → x-6-6-6-4-1	Eb7sus2/Bb → x-6-6-6-4-6	Eb7sus2/Db → x-6-6-6-2-x
E7sus2	0-2-4-4-0-3	E7sus2/Gb → 0-2-4-4-0-2	E7sus2/B → 0-2-4-4-0-7	E7sus2/D → 0-2-4-4-3-x
F7sus2	1-3-3-3-1-4	F7sus2/G → 1-3-3-3-1-3	F7sus2/C → 1-3-3-3-1-x	F7sus2/Eb → 1-3-3-3-4-x
Gb7sus2	2-4-4-4-2-5	Gb7sus2/Ab → 2-4-4-4-2-4	Gb7sus2/Db → 2-4-4-4-2-x	Gb7sus2/E → 2-4-4-4-2-0
G7sus2	3-0-0-0-1-1	G7sus2/A → 3-0-0-0-1-5	G7sus2/D → 3-0-0-0-3-x	G7sus2/F → 3-0-0-0-1-1
Ab7sus2	4-6-6-6-4-7	Ab7sus2/Bb → 4-6-6-6-4-6	Ab7sus2/Eb → 4-6-6-6-4-x	Ab7sus2/Gb → 4-6-6-6-4-2
A7sus2	x-0-2-0-0-0	A7sus2/B → x-0-2-0-0-7	A7sus2/E → x-0-2-0-0-0	A7sus2/G → x-0-2-0-0-3
Bb7sus2	x-1-3-1-1-1	Bb7sus2/C → x-1-3-1-1-x	Bb7sus2/F → x-1-3-1-1-1	Bb7sus2/Ab → x-1-3-1-1-4
B7sus2	x-2-4-2-2-2	B7sus2/Db → x-2-4-2-2-x	B7sus2/Gb → x-2-4-2-2-2	B7sus2/A → x-2-4-2-2-5

53. 7sus4 Inversions — Guitar

The 7sus4 chord (Root, P 4th, P 5th, Min 7th) inversions on guitar create increasing tension as the inversion number rises. The 1st inversion (4th in bass) sounds like a IV7 chord. The 3rd inversion (minor 7th in bass) is the most unresolved and desperate — it strongly wants to move to the tonic. These inversions map out the classic jazz resolution path: 7sus4 → 7 → I.

Root + 5 + 2 + 3 semitones | Root, P 4th, P 5th, Min 7th

Fret shape: x = mute | 0 = open | number = fret. Strings written: Str6 (Low E) → Str5 (A) → Str4 (D) → Str3 (G) → Str2 (B) → Str1 (High e)

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
C7sus4	x-3-3-3-1-1	C7sus4/F → x-3-3-3-1-1	C7sus4/G → x-3-3-3-1-3	C7sus4/Bb → x-3-3-3-1-6

Chord	Root Position — (Root = bass)	1st Inversion — (3rd = bass)	2nd Inversion — (5th = bass)	3rd Inversion — (7th = bass)
Db7sus4	x-4-4-4-2-2	Db7sus4/Gb → x-4-4-4-2-2	Db7sus4/Ab → x-4-4-4-2-4	Db7sus4/B → x-4-4-4-2-7
D7sus4	x-0-0-2-1-1	D7sus4/G → x-0-0-2-1-3	D7sus4/A → x-0-0-2-1-5	D7sus4/C → x-0-0-2-1-x
Eb7sus4	x-6-6-6-4-4	Eb7sus4/Ab → x-6-6-6-4-4	Eb7sus4/Bb → x-6-6-6-4-6	Eb7sus4/Db → x-6-6-6-2-x
E7sus4	0-2-2-2-3-0	E7sus4/A → 0-2-2-2-3-5	E7sus4/B → 0-2-2-2-3-7	E7sus4/D → 0-2-2-2-3-x
F7sus4	1-3-3-3-4-1	F7sus4/Bb → 1-3-3-3-4-6	F7sus4/C → 1-3-3-3-1-x	F7sus4/Eb → 1-3-3-3-4-x
Gb7sus4	2-4-4-4-5-2	Gb7sus4/B → 2-4-4-4-5-7	Gb7sus4/Db → 2-4-4-4-2-x	Gb7sus4/E → 2-4-4-4-5-0
G7sus4	3-3-0-0-1-1	G7sus4/C → 3-3-0-0-1-x	G7sus4/D → 3-3-0-0-3-x	G7sus4/F → 3-3-0-0-1-1
Ab7sus4	4-6-6-6-7-4	Ab7sus4/Db → 4-6-6-6-2-x	Ab7sus4/Eb → 4-6-6-6-4-x	Ab7sus4/Gb → 4-6-6-6-7-2
A7sus4	x-0-2-0-3-0	A7sus4/D → x-0-2-0-3-x	A7sus4/E → x-0-2-0-3-0	A7sus4/G → x-0-2-0-3-3
Bb7sus4	x-1-3-1-4-1	Bb7sus4/Eb → x-1-3-1-4-x	Bb7sus4/F → x-1-3-1-4-1	Bb7sus4/Ab → x-1-3-1-4-4
B7sus4	x-2-4-2-5-2	B7sus4/E → x-2-4-2-5-0	B7sus4/Gb → x-2-4-2-5-2	B7sus4/A → x-2-4-2-5-5

📝 Grade 2 Summary — Quick Reference

Tune

- Tone = a musical note with a specific pitch
- Starting Pitch = the first note of a song — find it by humming and testing
- Follow pitches: notes move UP (step/jump) or DOWN (step/jump)
- Speed/Slow/Gaps/Bends = Tempo, Rests, Glissando
- Write tones using: letter names (C D E...) or numbers (1 2 3...)

Rhythm

- Beat = steady pulse. Bar = group of beats.
- Note values: Whole=4 beats, Half=2, Quarter=1, Eighth=0.5
- 2/4: 2 beats/bar (march). 3/4: 3 beats/bar (waltz). 4/4: 4 beats/bar (pop). 6/8: 6 beats/bar (compound)
- Use a metronome. Always start slow. Gradually increase tempo.

Scales

- Major Scale: W W H W W W H — sounds happy
- Minor Scale: W H W W H W W — sounds sad
- Sharps (#) raise a note. Flats (b) lower a note.
- Circle of Fifths: shows sharps and flats for each key
- Transposition: same melody, different key
- Modulation: key change within a song

Chords

- Chord = 3+ notes played together
- Major Chord = Root + 4 semitones + 7 semitones
- Minor Chord = Root + 3 semitones + 7 semitones
- Primary Chords: I, IV, V
- Secondary Chords: ii, iii, vi, vii°
- Inversions: rearrange chord notes for smoother movement
- 7th Chords: add the 7th degree for jazz/blues colour

- Chord Progressions: I-V-vi-IV (pop), ii-V-I (jazz), 12-bar blues

🏆 **Final Message to Every Student**
Music is a language. Just like learning to read and write takes time, learning to hear and play music takes practice. But every day you practice, you get better. Keep listening. Keep playing. Keep exploring. You are a musician!

End of Grade 2 — Listen and Play

🎵 **Keep Practising** 🎵